

The factors responsible for this sterility are complex. For example, urine kills some bacteria due to its low pH and the presence of urea and other metabolic end products (uric acid, hippuric acid, indican, fatty acids, mucin, enzymes). The kidney medulla is so hypertonic that few organisms can survive. The lower urinary tract is flushed with urine and some mucus 4 to 10 times each day, eliminating potential pathogens. In males the anatomical length of the urethra (20 cm) provides a distance barrier that excludes microorganisms from the urinary bladder. Conversely the short urethra (5 cm) in females is more readily traversed by microorganisms; this explains why general urinary tract infections are 14 times more common in females than in males.

The vagina has another unique defense. Under the influence of estrogens, the vaginal epithelium produces increased amounts of glycogen that acid-tolerant *Lactobacillus acidophilus* bacteria called Döderlein's bacilli degrade to form lactic acid. Normal vaginal secretions contain up to 10^8 Döderlein's bacilli per ml. Thus an acidic environment (pH 3 to 5) unfavorable to most organisms is established. Cervical mucus also has some antibacterial activity.

The Eye

The conjunctiva is a specialized mucus-secreting epithelial membrane that lines the interior surface of each eyelid and the exposed surface of the eyeball. It is kept moist by the continuous flushing action of tears (lacrimal fluid) from the lacrimal glands. Tears contain large amounts of lysozyme, lactoferrin, and sIgA (*see p. 738*) and thus provide chemical as well as physical protection.

1. Describe the different antimicrobial defense mechanisms that operate within the respiratory system of mammals.
2. What factors operate within the gastrointestinal system that help prevent the establishment of pathogenic microorganisms?
3. Except for the anterior portion of the urethra, why is the genitourinary tract a sterile environment?

Chemical Barriers

Mammalian hosts have a chemical arsenal with which to combat the continuous onslaught of microorganisms. Some of these chemicals (gastric juices, salivary glycoproteins, lysozyme, oleic acid on the skin, urea) have already been discussed with respect to the specific body site(s) they protect. In addition, blood, lymph, and other body fluids contain a potpourri of defensive chemicals such as bacteriocins, beta-lysin, and other polypeptides.

Bacteriocins

As previously noted, the first line of defense against microorganisms is the host's anatomical barrier, consisting of the skin and mucous membranes. These surfaces are colonized by normal microbiota, which by themselves provide a biological barrier against uncontrolled proliferation of foreign microorganisms. Many of these normal bacteria synthesize and release plasmid-

encoded toxic proteins (e.g., colicin, staphylococcin) called **bacteriocins** that are lethal to related species. Bacteriocins may give their producers an adaptive advantage against other bacteria. Sometimes they may increase bacterial virulence by damaging host cells such as mononuclear phagocytes.

Most bacteriocins that have been identified are peptides or proteins and are produced by gram-negative bacteria. (However, recently it has been discovered that some gram-positive bacteria produce bacteriocin-like peptides). For example, *E. coli* synthesizes bacteriocins called **colicins**, which are coded for by several different plasmids (ColB, ColE1, ColE2, ColI, and ColV). Some colicins bind to specific receptors on the cell envelope of sensitive target bacteria and cause cell lysis, attack specific intracellular sites such as ribosomes, or disrupt energy production. It is now widely recognized that these antimicrobial peptides act as defensive effector molecules in the large intestine.

Beta-Lysin and Other Polypeptides

Beta-lysin is a cationic polypeptide released from blood platelets; it can kill some gram-positive bacteria by disrupting their plasma membranes. Other cationic polypeptides include leukins, plakins, cecropins, and phagocytin. A zinc-containing polypeptide, known as the prostatic antibacterial factor, is an important antimicrobial substance secreted by the prostate gland in males.

1. How does beta-lysin function against gram-positive bacteria?
2. How do bacteriocins function?

31.6 Inflammation

Inflammation [Latin, *inflammatio*, to set on fire] is an important nonspecific defense reaction to tissue injury, such as that caused by a pathogen or wound. Acute inflammation is the immediate response of the body to injury or cell death. The gross features were described over 2,000 years ago and are still known as the cardinal signs of inflammation. These signs include redness (*rubor*), warmth (*calor*), pain (*dolor*), swelling (*tumor*), and altered function (*functio laesa*).

The acute inflammatory response begins when injured tissue cells release chemical signals (inflammatory mediators) that activate the inner lining (endothelium) of nearby capillaries (**figure 31.11**). Within the capillaries **selectins** (a family of cell adhesion molecules) are displayed on the activated endothelial cells—first P-selectin and then E-selectin. These adhesion molecules randomly attract and attach neutrophils to the endothelial cells, slow the neutrophils down, and cause them to roll along the endothelium. As the neutrophils (**figure 31.3**) roll along the endothelium, they encounter the inflammatory mediators that act as activating signals. These signals activate **integrins** (adhesion receptors) on the neutrophils. The neutrophil integrins then tightly attach to endothelial adhesion molecules such as the intracellular adhesion molecule-1 (ICAM-1) and vascular cell adhesion molecule-1 (VCAM-1). This causes the

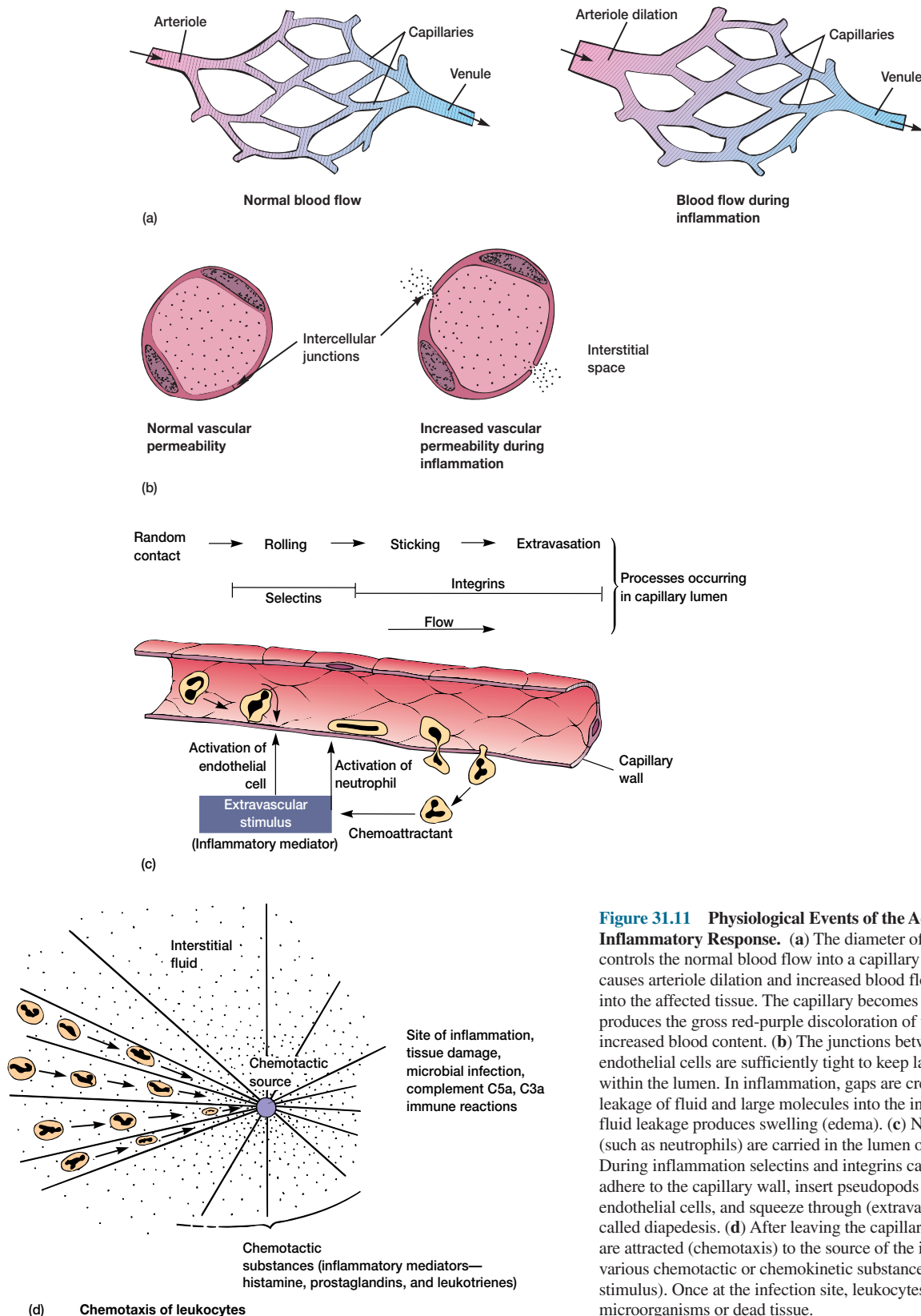


Figure 31.11 Physiological Events of the Acute Inflammatory Response.

(a) The diameter of the arteriole controls the normal blood flow into a capillary bed. Inflammation causes arteriole dilation and increased blood flow (hyperemia) into the affected tissue. The capillary becomes distended and produces the gross red-purple discoloration of tissue due to an increased blood content. (b) The junctions between capillary endothelial cells are sufficiently tight to keep large molecules within the lumen. In inflammation, gaps are created that allow leakage of fluid and large molecules into the interstitial space. The fluid leakage produces swelling (edema). (c) Normally leukocytes (such as neutrophils) are carried in the lumen of a capillary. During inflammation selectins and integrins cause leukocytes to adhere to the capillary wall, insert pseudopods between the endothelial cells, and squeeze through (extravasation) by a process called diapedesis. (d) After leaving the capillary, the leukocytes are attracted (chemotaxis) to the source of the inflammation by various chemotactic or chemokinetic substances (extravascular stimulus). Once at the infection site, leukocytes phagocytose the microorganisms or dead tissue.

neutrophils to stick to the endothelium and stop rolling. The neutrophils now undergo dramatic shape changes, squeeze through the endothelial wall (extravasation) into the interstitial tissue fluid, migrate to the site of injury, and attack the pathogen or other cause of the tissue damage. Neutrophils and other leukocytes are attracted to the infection site by chemotactic factors or chemotaxins such as substances released by bacteria and mast cells, and tissue breakdown products. Depending on the severity and nature of tissue damage, other types of leukocytes (e.g., lymphocytes, monocytes, and macrophages) may follow the neutrophils.

The inflammatory mediators that are released by the injured tissue cells also raise the acidity in the surrounding extracellular fluid. This decrease in pH activates the extracellular enzyme kallikrein, which splits bradykinin from its long precursor chain. Bradykinin binds to receptors on the capillary wall, opening the junctions between cells and allowing fluid and infection-fighting leukocytes to leave the capillary and enter the infected tissue. Simultaneously bradykinin (**figure 31.12**) binds to mast cells in the connective tissue associated with most small blood vessels. This activates the mast cells by causing an influx of calcium ions, which leads to degranulation and release of preformed mediators such as histamine. If nerves in the infected area are damaged, they release substance P, which also binds to mast cells, boosting preformed-mediator release. Histamine in turn makes the intercellular junctions in the capillary wall wider so that more fluid, leukocytes, kallikrein, and bradykinin precursors move out, causing edema. Bradykinin then binds to nearby capillary cells and stimulates the production of prostaglandins (PGE_2 and $\text{PGF}_{2\alpha}$) to promote tissue swelling in the infected area. Prostaglandins also bind to free nerve endings, making them fire and start a pain impulse.

The change in mast cell plasma membrane permeability associated with activation allows phospholipase A2 to release arachidonic acid. Arachidonic acid is then metabolized by the cyclooxygenase or lipoxygenase pathways, depending on mast cell type. The newly synthesized mediators include prostaglandins E_2 and $\text{F}_{2\alpha}$, thromboxane A_2 , slow-reacting substance (SRS), and leukotrienes ($\text{LTC}_4 + \text{LTD}_4$). All play various roles in the inflammatory response.

During acute inflammation, the offending pathogen is neutralized and eliminated by a series of events, the most important of which are the following:

1. The increase in blood flow and capillary dilation bring into the area more antimicrobial factors and leukocytes that destroy the pathogen. Dead cells also release antimicrobial factors.
2. The rise in temperature stimulates the inflammatory response and may inhibit microbial growth.
3. A fibrin clot often forms and may limit the spread of the invaders so that they remain localized.
4. Phagocytes collect in the inflamed area and phagocytose the pathogen. In addition, chemicals stimulate the bone marrow to release neutrophils and increase the rate of granulocyte production.

Chronic Inflammation

Chronic inflammation is a slow process characterized by the formation of new connective tissue, and it usually causes permanent tissue damage. Superficially, the difference between acute and chronic inflammation is one of duration. Regardless of the cause, chronic inflammation lasts two weeks or longer. Chronic inflammation can occur as a distinct process without much acute inflammation. The persistence of bacteria by a variety of mechanisms can stimulate chronic inflammation. For example, the mycobacteria have cell walls with a very high lipid and wax content, making them relatively insensitive to phagocytosis. The bacteria that cause tuberculosis, leprosy, and syphilis often survive within the macrophage. In addition, some bacteria produce toxins that stimulate tissue-damaging reactions even after bacterial death.

Chronic inflammation is characterized by a dense infiltration of lymphocytes and macrophages. If the macrophages are unable to protect the host from tissue damage, the body attempts to wall off and isolate the site by forming a **granuloma** [Latin, *granulum*, a small particle; Greek *oma*, to form]. Granulomas are formed when neutrophils and macrophages are unable to destroy the microorganism during inflammation. Infections caused by some bacteria (listeriosis, brucellosis), fungi (histoplasmosis, coccidioidomycosis), helminth parasites (leishmaniasis, schistosomiasis), and large antibody-antigen complexes (rheumatoid arthritis) result in granuloma formation and chronic inflammation.

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1. What major events occur during an inflammatory reaction, and how do they contribute to pathogen destruction?
 2. What causes degranulation of mast cells?
 3. How does chronic inflammation differ from acute inflammation?
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31.7 The Complement System

Complement was discovered many years ago as a heat-labile component of human blood plasma that augments opsonization (section 31.8) of bacteria by antibodies and helps other antibodies kill bacteria. This activity was said to “complement” the antibacterial activity of antibody; hence, the name complement. It is now known that the **complement system** is composed of a large number of serum proteins that play a major role in the animal’s defensive immune response. For example, complement proteins can lyse antibody-coated eucaryotic cells and bacteria (cytolysis). Complement can mediate inflammation and attract and activate phagocytic cells. Generally, complement proteins amplify the effects of antibodies (e.g., lysis of cells).

The complement cascade is made up of at least 20 complement proteins designated C1 (which has three protein subcomponents) through C9 in addition to Factor B, Factor $\bar{\text{D}}$, Factor H, Factor I, C4b binding protein, C1 INH complex, S protein, and

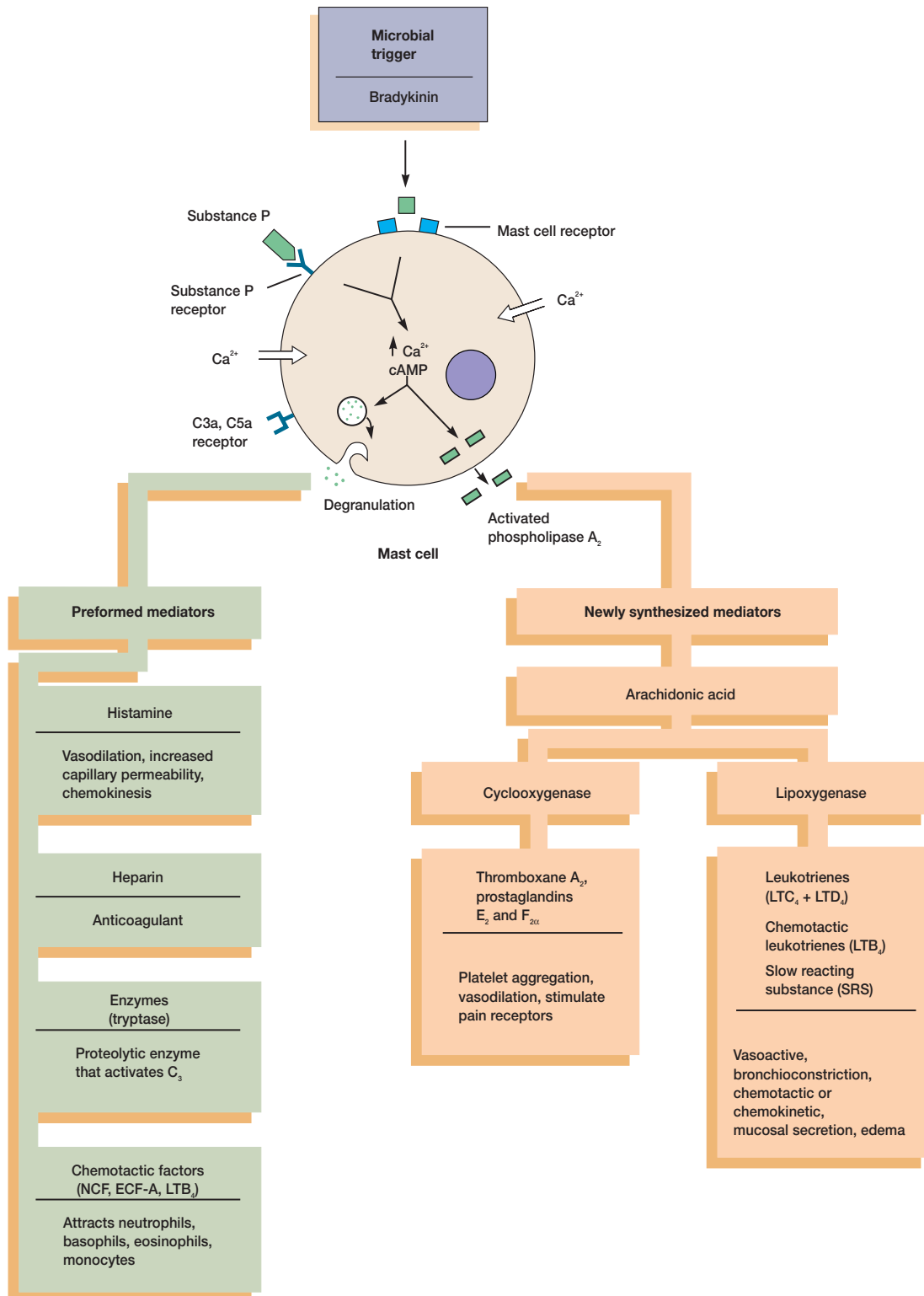


Figure 31.12 Biochemical Events of Inflammation. Mast cell activation and physiological effects of mast cell-derived performed and newly synthesized mediators that lead to the inflammatory response.

Table 31.1 Proteins of the Complement Cascade in Serum

Protein	Fragment	Function
Recognition Unit		
C1	q	Binds to the Fc portion of antigen-antibody complexes
	r	Subunit of C1; activates C1s
	s	Cleaves C4 and C2 due to its enzymatic activity
Activation Unit		
C2		Causes viral neutralization
C3	a	Anaphylatoxin, immunoregulatory
	b	Key component of the alternative pathway and major opsonin in serum
	e	Induces leukocytosis
C4	a	Anaphylatoxin
	b	Causes viral neutralization; opsonin
Membrane Attack Unit		
C5	a	Anaphylatoxin; principal chemotactic factor in serum; induces neutrophil attachment to blood vessel walls
	b	Initiates membrane attack
C6	}	Participate with C5b in formation of the membrane attack complex that lyses targeted cells
C7		
C8		
C9		
Alternative Pathway		
Factor B		Causes macrophage spreading on surfaces; precursor of C3 convertase
Factor D̄		Cleaves Factor B to form active C3Bb in alternative pathway
Properdin		Stabilizes alternative pathway C3 convertase
Regulatory Proteins		
Factor H		Promotes C3b breakdown and regulates alternative pathway
Factor I		Degrades C3b and regulates alternative pathway
C4b binding protein		Inhibits assembly and accelerates decay of C4bC2a
C1 INH complex		Binds to and dissociates C1r and C1s from C1
S protein		Binds fluid-phase C5b67; prevents membrane attachment

properdin (**table 31.1**). The complement system acts in a cascade fashion, the activation of one component resulting in the activation of the next. Collectively the complement proteins make up much of the globulin fraction of serum (*see section 32.3 and figure 32.6*). Within plasma and other body fluids, complement proteins are in an inactive state.

There are three pathways of complement activation: the classical, alternative, and lectin pathways. Although they employ similar mechanisms, specific proteins are unique to the first part of each pathway. Since the classical pathway involves an understanding of the specific immune response, it will be discussed in section 32.7.

The **alternative complement pathway** (**figure 31.13**) plays an important role in the innate, nonspecific immune defense against intravascular invasion by bacteria and some fungi. The alternative pathway begins with cleavage of C3 into fragments C3a and C3b by a blood enzyme. These fragments are produced at a slow rate and do not carry out the next step, the cleavage of C5, because free C3b is rapidly cleaved into inactive fragments. However, C3b becomes stable when it binds to the lipopolysaccharide (LPS) of gram-negative bacterial cell walls, to aggregates of IgA or IgE, or to some endotoxins. A protein in blood termed Factor B adsorbs to bound C3b and is cleaved into two fragments by Factor $\bar{D}$, leading to the formation of active enzyme C3bBb (this complex is sometimes called the C3 convertase because it cleaves more C3 to C3a and C3b). C3bBb is further stabilized by a second blood protein, properdin, and is changed into the C5 convertase (C3bBb3b). The convertase then cleaves C5 to C5a and C5b. C6 and C7 rapidly bind to C5b, forming a C5b67 complex that possesses an unstable membrane-binding site; once bound to a membrane, this complex is stable. C8 and C9 then bind, forming the **membrane attack complex** (C5b6789) that creates a pore in the plasma membrane of the target cell (**figure 31.14**). It is believed that the actual pore is a doughnut-shaped polymer of C9. If the cell is eucaryotic, Na⁺ and H₂O enter through the pore and the cell lyses osmotically. Lysozyme can pass through pores in the outer membrane of gram-negative cell walls and digest the peptidoglycan cell wall, thus weakening the wall and aiding lysis. In contrast, gram-positive bacteria resist the cytolytic action of the membrane attack complex because they lack an exposed outer membrane and have a thick peptidoglycan that prevents an attack on the plasma membrane. However, gram-positive bacteria are very susceptible to opsonization (**figure 31.15**) when coated with C3b.

The **lectin complement pathway** activates C3 convertase by using a lectin, a special protein that binds to specific carbohydrates. When macrophages ingest viruses, bacteria, and other foreign material, they release chemicals that stimulate liver cells to secrete mannose-binding protein (MBP). Because mannose is a major component of bacterial cell walls and of some virus envelopes and immune complexes, MBP binds to many pathogens and soluble immune complexes. MBP is an opsonin and directly enhances phagocytosis. The lectin also activates complement in two ways (**figure 31.13**). After binding to the surface, it can trigger the alternative pathway directly. In addition, if MBP is bound to the MBP-associated serine esterase (MASP), it activates the classical complement pathway. Thus although the lectin pathway activates the complement cascade just as the classical and alternative pathways do, it uses a mechanism that is independent of antibody-antigen interactions (the classical pathway) and the interaction of complement with pathogen surfaces (alternative pathway).

This overview of the alternative and lectin complement pathways provides a basis for consideration of the function of complement as an integrated system during an animal's defensive effort. Gram-negative bacteria arriving at a local tissue site will interact with components of the alternative pathway, resulting in the generation of biologically active fragments, opsonization of the bacteria, and initiation of the lytic sequence. If the bacteria

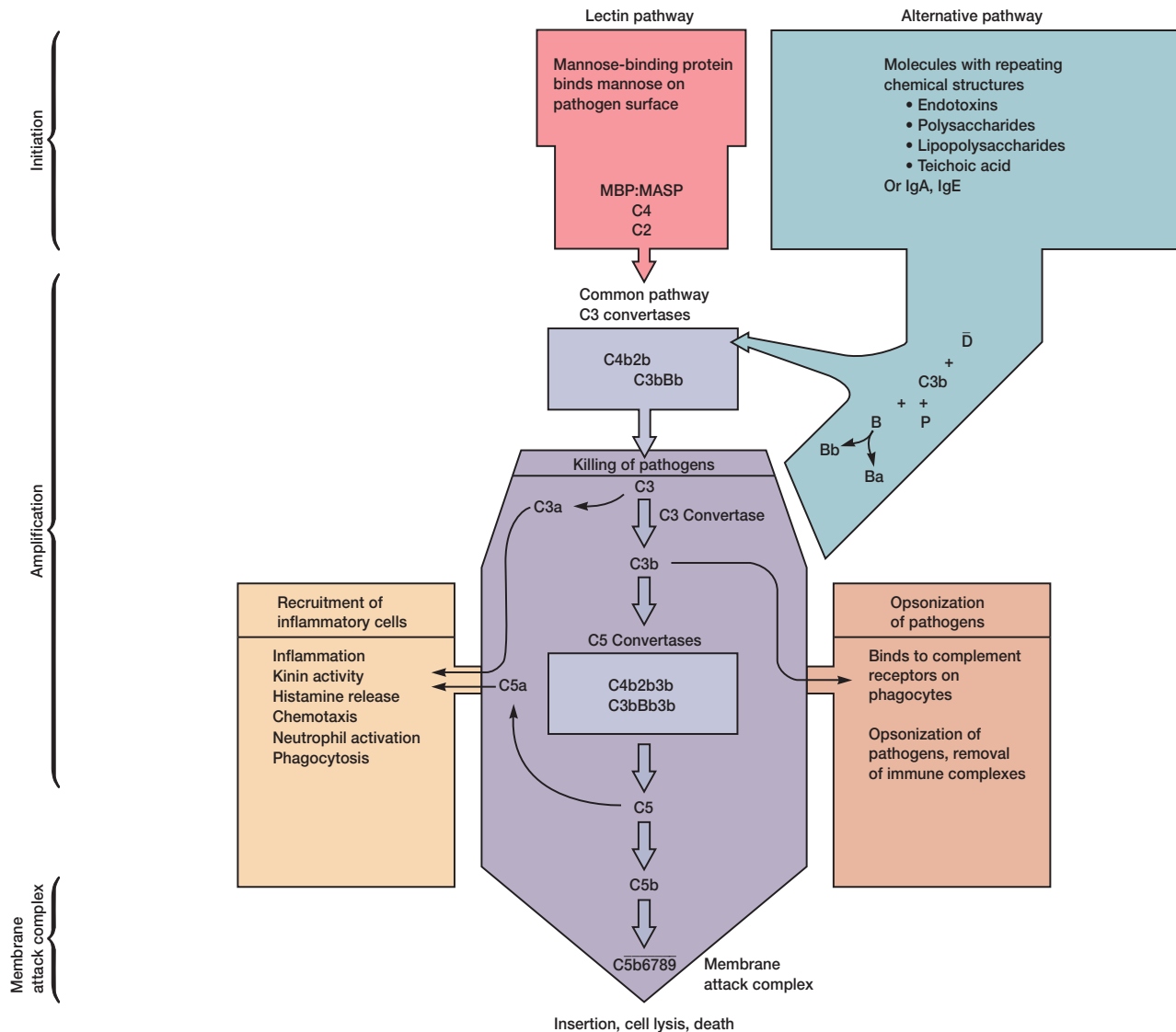


Figure 31.13 Alternative and Lectin Complement Pathways. Within each pathway the components are arranged in order of their activation and aligned opposite their functional and structural analogs in the opposite pathway. Two of the three pathways of complement activation are shown: (1) the lectin-mediated pathway (red), which is triggered by a serum protein that binds to mannose residues of bacteria; and (2) the alternative pathway (blue), which is triggered directly on pathogen surfaces. Both of these generate a crucial enzymatic activity (common pathway of C3 convertases) that in turn generates the effector activity of complement. The three main consequences of complement activation are recruitment of inflammatory cells (left box), direct killing of pathogens by the membrane attack complex (middle box), and opsonization of pathogens (right box).

persist or if they invade the animal a second time, antibody responses also will activate the classical pathway (*see section 32.7*).

The generation of complement fragments C3a and C5a leads to several important inflammatory effects. Mast cells release their contents, and the blood supply to the area increases markedly (hyperemia) as blood vessels dilate due to released histamine (figure

31.12). These fragments also cause the release of neutrophils from the bone marrow into the circulation. Neutrophils then make their way to the site of hyperemia where, in the presence of C5a, they attach to the endothelium and leave the blood. C5a induces a directed, chemotactic migration of neutrophils to the site of complement activation. Macrophages in the area can synthesize even

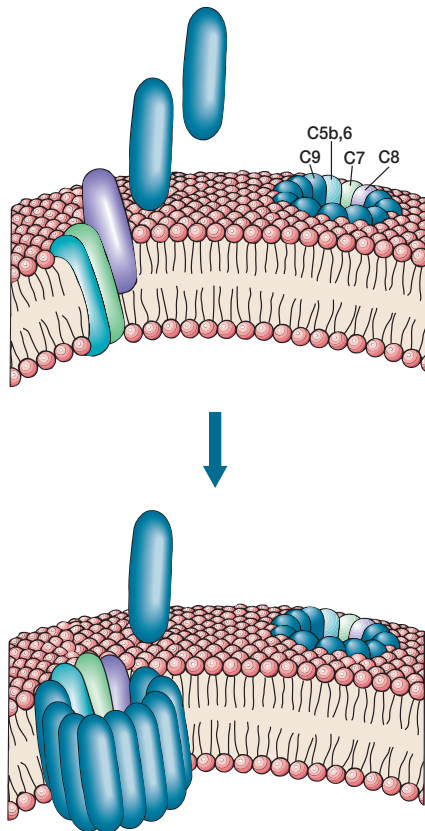


Figure 31.14 The Membrane Attack Complex. The membrane attack complex (MAC) is a tubular structure that forms a transmembrane pore in the target cell's plasma membrane. This representation is the subunit architecture of the membrane attack complex. The transmembrane channel is formed by a C5b678 complex and 10 to 16 polymerized molecules of C9.

more complement components to interact with the bacteria. All these defensive events promote the ingestion and ultimate destruction of the bacteria by the neutrophils and macrophages.

1. List the major functions of complement.
2. How is the alternative pathway activated? The lectin pathway?
3. What is the membrane attack complex, and how does its formation lead to cell lysis?
4. What role do complement fragments C3a and C5a play in an animal's defense against gram-negative bacteria?

31.8 Phagocytosis

During their lifetimes, humans and other vertebrates encounter many microbial species, but only a few of these species can grow and cause serious disease in otherwise healthy hosts.

Phagocytic cells (monocytes, tissue macrophages, and neutrophils) are an important early defense against invading microorganisms. These phagocytic cells recognize, ingest, and kill many extracellular microbial species by the process called **phagocytosis** [Greek *phagein*, to eat, and *cyte*, cell, and *osis*, a process]. The concept of phagocytosis was briefly introduced in section 4.5 within the discussion of lysosomes and how certain cells obtain nutrients by endocytosis. Phagocytosis will now be considered in more detail in the context of nonspecific host resistance.

Phagocytic cells use two basic molecular mechanisms for the recognition of microorganisms: (1) opsonin-dependent (opsonic) and (2) opsonin-independent (nonopsonic) recognition. The phagocytic process can be greatly enhanced by opsonization. **Opsonization** [Greek *opson*, to prepare victims for] is a process in which microorganisms or other particles are coated by serum components (antibodies and/or complement C3b) thereby preparing them for recognition and ingestion by phagocytic cells. In the **opsonin-dependent** recognition mechanism, it is the serum components that function as a bridge between the microorganisms and the phagocyte. They act by binding to the surface of the microorganism at one end and to specific receptors on the phagocyte surface at the other (figure 31.15b–d).

The **opsonin-independent** mechanism does not require opsonins and uses other nonspecific (figure 31.15a) and specific receptors (figure 31.16a) on phagocytic cells that recognize structures expressed on the surface of different microorganisms. Three main forms of recognition in opsonin-independent phagocytosis have been identified (table 31.2). One mode, termed lectin phagocytosis, is based on the recognition between surface lectins on one cell and surface carbohydrates on the opposing cell. The second mode is the result of protein-protein interactions between the Arg-Gly-Asp peptide sequence of microorganisms and macrophage receptors. Hydrophobic interactions between bacteria and phagocytic cells also promote phagocytosis. It should be noted that a particular microbial species can express multiple adhesins, each recognized by a distinct receptor present on phagocytic cells.

Once ingested by phagocytosis, membrane-enveloped microorganisms are delivered to a lysosome by fusion of the phagocytic vacuole, called a **phagosome**, with the lysosome membrane, forming a new vacuole called a **phagolysosome** (figure 31.16b). Lysosomes contribute to the phagolysosome a variety of hydrolases such as lysozyme, phospholipase A₂, ribonuclease, deoxyribonuclease, and proteases. An acidic vacuolar pH favors the activity of the hydrolases. Collectively these participate in the destruction of the entrapped microorganism.

Besides these oxygen-independent lysosomal hydrolases, macrophage and neutrophil lysosomes contain oxygen-dependent enzymes that can produce toxic **reactive oxygen intermediates (ROIs)** such as the superoxide radical ($O_2^{\cdot-}$), hydrogen peroxide (H_2O_2), singlet oxygen (1O_2), and hydroxyl radical ($OH\cdot$). The NADPH required for this process is supplied by a large increase in pentose phosphate pathway activity (see figure 9.6). Neutrophils also contain myeloperoxidase and produce

Phagocytic cell	Degree of binding	Opsonin
 (a) Attachment by nonspecific receptors	±	-
 (b)	+	Antibody
 (c)	++	Complement C3b
 (d)	++++	Antibody and complement C3b

Figure 31.15 Opsonization. (a) A phagocytic cell has some intrinsic ability to bind directly to a microorganism through nonspecific receptors. (b) This binding ability is enhanced if the microorganism elicits the formation of antibodies (Ab) that act as a bridge to attach the microorganism to the Fc receptor on the phagocytic cell. (c) If the microorganism has activated complement (C3b), the degree of binding is further enhanced by the C3b receptor. (d) If both antibody and C3b opsonize, binding is greatly enhanced.

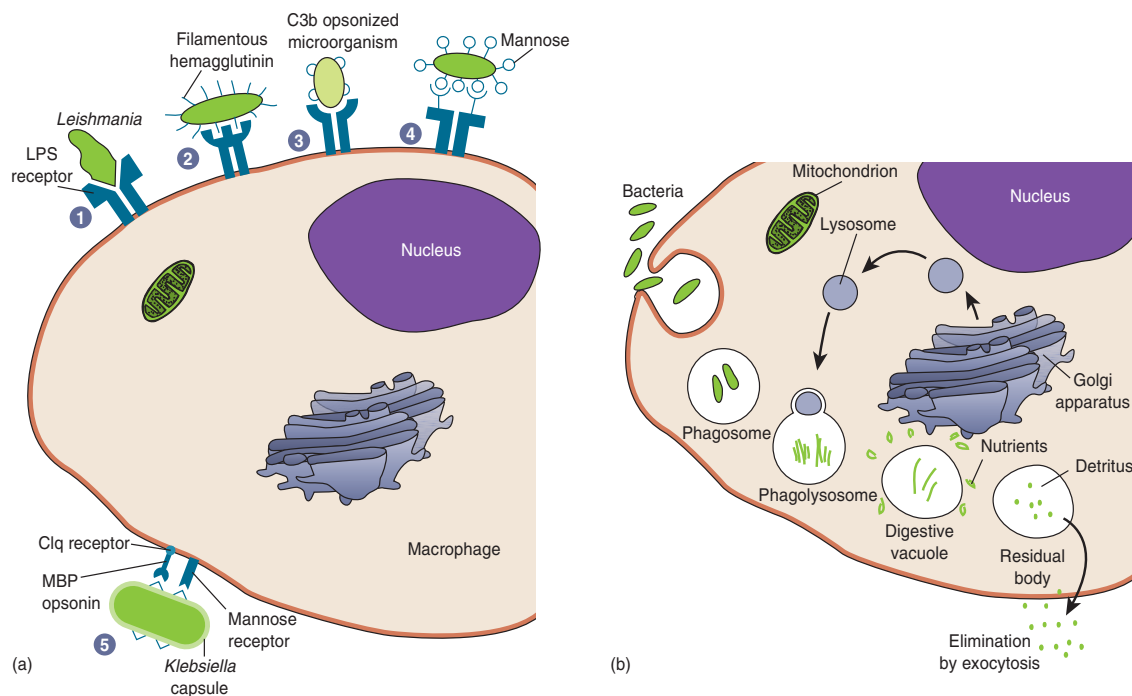


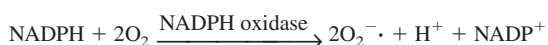
Figure 31.16 Phagocytosis. (a) Drawing shows receptors on a phagocytic cell, such as a macrophage, and the corresponding adhesins on microbial surfaces participating in phagocytosis. (1) The LPS binding site for lipophosphoglycan of *Leishmania* spp; (2) filamentous hemagglutinin of *Bordetella pertussis*; (3) binding to C3b on an opsonized cell; (4) the mannose-containing oligosaccharide side chain for lectin phagocytosis mediated by type 1 fimbriated bacteria; and (5) the participation of capsular polysaccharide of *Klebsiella pneumoniae* in nonopsonic binding mediated by the mannose receptor. (b) Drawing of phagocytosis, showing ingestion, intracellular digestion, and exocytosis.

Table 31.2 Nonopsonic Modes of Recognition by Phagocytes

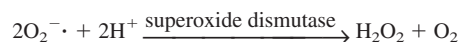
Type of Interaction	Bacterial Ligand (and Example)	Phagocytic Receptor (and Example)
Lectin-carbohydrate	Lectin (Type I fimbriae; <i>see figure 3.30</i>) Polysaccharide (capsule; <i>see figure 3.27</i>)	Glycoprotein (integrins) Lectin (Man/GlcNAc receptors)
Protein-protein	Arginine-glycine-aspartic acid; RGD-containing proteins (filamentous hemagglutinin; <i>see table 34.3</i>)	RGC receptor (integrins)
Hydrophobic protein	Glycolipid (lipoteichoic acid; <i>see figure 3.21</i>)	Lipid receptors (integrins)

hypochlorous acid. Some reactions forming these toxic products are shown here.

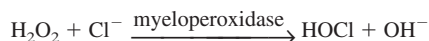
Superoxide formation



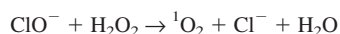
Hydrogen peroxide formation



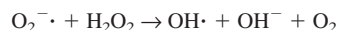
Hypochlorous acid formation



Singlet oxygen formation



Hydroxyl radical formation



These reactions result from the **respiratory burst** that accompanies the increased oxygen consumption and ATP generation needed for phagocytosis. Because these reactions occur as soon as the phagosome is formed, lysosome fusion is not necessary for the respiratory burst. The toxic oxygen products produced are effective in killing invading microorganisms. [Oxygen and microbial growth \(pp. 127–29\)](#)

Recently macrophages, neutrophils, and mast cells have been shown to form **reactive nitrogen intermediates (RNIs)**. These molecules include nitric oxide (NO) and its oxidized forms, nitrite (NO_2^-) and nitrate (NO_3^-). The RNIs are very potent cytotoxic agents, and may be either released from cells or generated within cell vacuoles. Nitric oxide is probably the most effective RNI. Macrophages produce it from the amino acid arginine. Nitric oxide can block cellular respiration by complexing with the iron in electron transport proteins. Macrophage destruction of the herpes simplex virus, the protozoa *Toxoplasma gondii* and *Leishmania major*, the opportunistic fungus *Cryptococcus neoformans*, the metazoan pathogen *Schistosoma mansoni*, and tumor cells involves RNIs.

Neutrophil granules also contain a variety of other microbicidal substances such as several cationic proteins, the bactericidal permeability increasing protein (BPI), and the family of broad-

spectrum antimicrobial peptides called **defensins**. There are four human defensins, called human neutrophil proteins (HNPs): HNP-1, 2, 3, and 4. These defensins are synthesized by myeloid precursor cells during their sojourn in the bone marrow, and are then stored in the cytoplasmic granules of mature cells. This compartmentation strategically locates defensins for extracellular secretion or delivery to phagocytic vacuoles. Susceptible microbial targets include a variety of gram-positive and gram-negative bacteria, yeasts and molds, and some viruses. Defensins act against bacteria and fungi by permeabilizing cell membranes. They form voltage-dependent membrane channels that allow ionic efflux. Antiviral activity involves direct neutralization of enveloped viruses; nonenveloped viruses are not affected by defensins.

1. What is the difference between opsonic and nonopsonic phagocytosis?
2. Once a phagolysosome forms, how is the entrapped microorganism destroyed?
3. What is the purpose of the respiratory burst that occurs within macrophages? Describe the nature and function of reactive oxygen and nitrogen intermediates.
4. What are defensins? How do they function?

31.9 Cytokines

Defense against viruses, microorganisms and their products, parasites, and cancer cells is mediated by both nonspecific and specific immunity. Cytokines are required for immunoregulation of both of these immune responses.

The term **cytokine** [Greek *cyto*, cell, and *kinesis*, movement] is a generic term for the soluble protein or glycoprotein released by one cell population that acts as an intercellular (between cells) mediator or signaling molecule. When released from mononuclear phagocytes, these proteins are called **monokines**; when released from T lymphocytes they are called **lymphokines**; when produced by a leukocyte and the action is on another leukocyte, they are **interleukins**; and if their effect is to stimulate the growth and differentiation of immature leukocytes in the bone marrow, they are called **colony-stimulating factors (CSFs)**. Recently cy-

Table 31.3 The Four Cytokine Families

Family	Examples	Functions
Chemokines	IL-8, RANTES, MIP (macrophage inflammatory protein)	Cytokines that are chemotactic and chemokinetic for leukocytes. They stimulate cell migration and attract phagocytic cells and lymphocytes. Chemokines play a central role in the inflammatory response.
Hematopoietins	Epo (erythropoietin), various colony-stimulating factors	Cytokines that stimulate and regulate the growth and differentiation processes involved in blood cell formation (hematopoiesis).
Interleukins	IL-1 to IL-18	Cytokines produced by lymphocytes and monocytes that regulate the growth and differentiation of other cells, primarily lymphocytes and hematopoietic stem cells. They often also have other biological effects.
Tumor necrosis factor (TNF) family	TNF- α , TNF- β , Fas ligand	Cytokines that are cytotoxic for tumor cells and have many other effects such as promoting inflammation, fever, and shock; some can induce apoptosis.

tokines have been grouped into the following categories or families: chemokines, hematopoietins, interleukins, and members of the tumor necrosis factor (TNF) family. Some examples of these cytokine families are provided in **table 31.3**.

Cytokines can affect the same cell responsible for their production (an autocrine function), nearby cells (a paracrine function), or can be distributed by the circulatory system to their target cells (an endocrine function). Their production is induced by nonspecific stimuli such as a viral, bacterial, or parasitic infection; cancer; inflammation; or the interaction between a T cell and antigen. Some cytokines also can induce the production of other cytokines.

Knowledge of the biological actions of cytokines has grown enormously over the past two decades. This is due in part to the incredible potency and range of effects on eucaryotic cells exhibited by cytokines, and to the fact that they are involved in all aspects of disease. Cytokines produce biological actions only when they act as ligands and bind to specific, high-affinity receptors called CDs (cell-associated differentiation antigens (*see section 32.2*) on the surface of target cells. The affinity of cytokine receptors for their ligands is very high, and consequently cytokines are effective at concentrations of 10^{-10} to 10^{-15} M. Most cells have relatively few receptors, hundreds to a few thousand, and a maximal cellular response results when only a small number of these are occupied by the cytokine. This binding activates specific intracellular signaling pathways and switches on genes that encode proteins essential to the appropriate target cell functions. Examples of these proteins include other cytokines, cell-to-cell adhesion receptors, proteases, lipid-synthesizing enzymes, and nitric oxide synthase. Cytokines can activate cell proliferation and/or cell differentiation (**figure 31.17**). They also can inhibit cell division and cause apoptosis (programmed cell death). Chemokines, one type of cytokine, stimulate chemotaxis and chemokinesis (i.e., they direct cell movement) and thus play an important role in the acute inflammatory response (*figure 31.11*). Some examples of important cytokines and their functions are given in **table 31.4**.

Interferons

Interferons (IFNs) are a group of related low molecular weight, regulatory cytokines produced by many eucaryotic cells in response to numerous inducers: a virus infection, double-

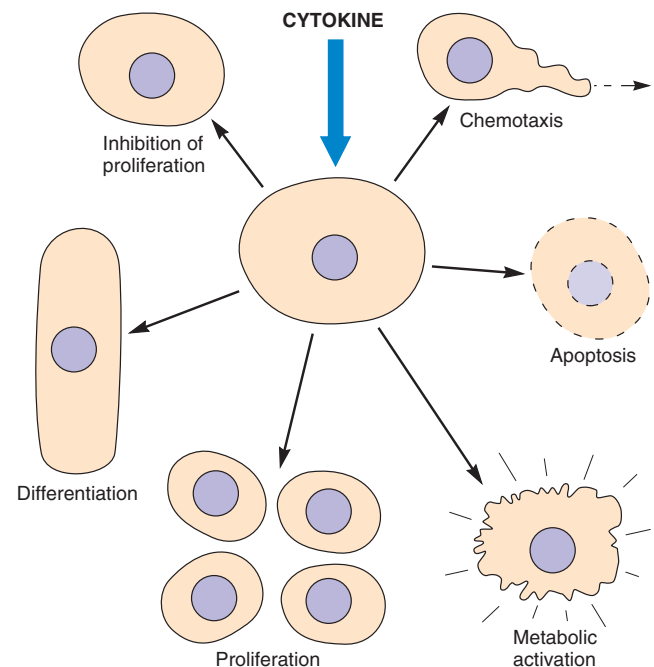


Figure 31.17 Range of Biological Actions That Cytokines Have on Eucaryotic Cells. Chemokines are one family of cytokines that induce leukocyte chemotaxis and migration. Other cytokines activate cell metabolism and synthesis. This can lead to the synthesis of a wide range of proteins including cyclooxygenase II, proteolytic enzymes, NO synthase, and various adhesion receptors. In addition, other cytokines can cause proliferation, inhibition of cell proliferation, or apoptosis.

stranded RNA, endotoxins, antigenic stimuli, mitogenic (stimulating mitosis) agents, and many pathogenic organisms capable of intracellular growth (*Listeria monocytogenes*, chlamydiae, rickettsias, protozoa). Interferons usually are species specific but virus nonspecific. Several classes of interferons are recognized: IFN- α is a family of 20 different molecules that can be synthesized by virus-infected leukocytes (*table 31.4*). IFN- β is derived from virus-infected fibroblasts; and IFN- γ is produced by antigen-stimulated T cells. Probably

Table 31.4 Some of the Cytokines That Mediate Immune Responses

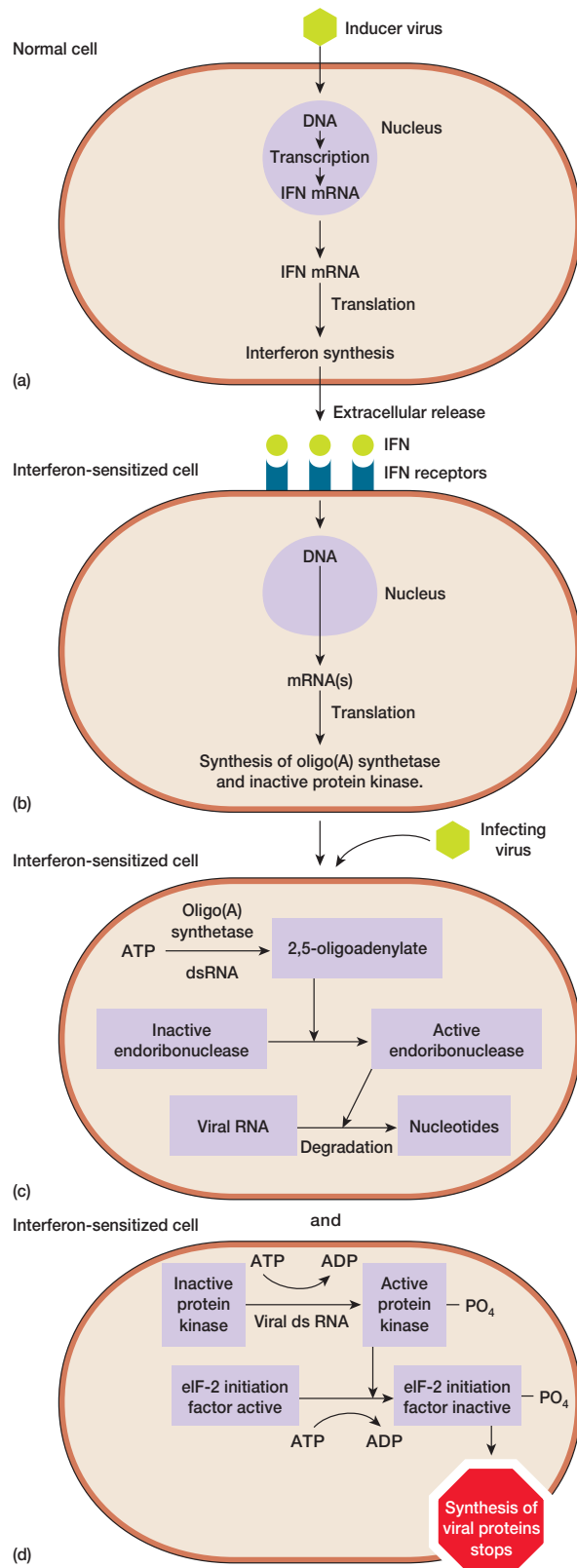
Cytokine	Cell Source	Functions
IL-1 (interleukin-1)	Monocytes/macrophages, endothelial cells, fibroblasts, neuronal cells, glial cells, keratinocytes, epithelial cells	Produces a wide variety of effects on the differentiation and function of cells involved in inflammatory and immune responses; also affects central nervous and endocrine systems; it is an endogenous pyrogen
IL-2 (interleukin-2, T-cell growth factor)	T cells (T _H 1)	Stimulates T-cell proliferation and differentiation, enhances cytolytic activity of NK cells, promotes proliferation and immunoglobulin secretion of activated B cells
IL-3 (interleukin-3)	T cells, keratinocytes, neuronal cells, mast cells	Stimulates the production and differentiation of macrophages, neutrophils, eosinophils, basophils, mast cells
IL-4 (interleukin-4, B-cell growth factor-1 [BCGF-1], B-cell stimulatory factor-1 [BCSF-1])	T cells (T _H 2), macrophages, mast cells, basophils, B cells	Induces the differentiation of naive CD4 ⁺ T cells into T-helper-like cells; induces the proliferation and differentiation of B cells; exhibits diverse effects on T cells, monocytes, granulocytes, fibroblasts, endothelial cells
IL-5 (interleukin-5)	T cells (T _H 2)	Growth and activation of B cells and eosinophils; activation of eosinophil function; chemotactic for eosinophils
IL-6 (interleukin-6, cytotoxic T-cell differentiation factor, B-cell differentiation factor)	T _H 2 cells, monocytes/macrophages, fibroblasts, hepatocytes, endothelial cells, neuronal cells	Activates hematopoietic cells; induces growth of T cells, B cells, hepatocytes, keratinocytes, and nerve cells; stimulates the production of acute-phase proteins
IL-8 (interleukin-8)	Monocytes, endothelial cells, fibroblasts, alveolar epithelium, T cells, keratinocytes, neutrophils, hepatocytes	Chemoattractant for PMNs; causes degranulation and expression of receptors; inhibits adhesion of PMNs to endothelium; promotes migration of PMNs through endothelium; induces adhesion of neutrophils to endothelial cells
IL-10 (interleukin-10)	T cells (T _H 2), B cells, macrophages, keratinocytes	Reduces the production of IFN- γ , IL-1, TNF- α , and IL-6 by macrophages; in combination with IL-3 and IL-4, causes mast cell growth; in combination with IL-2, causes growth of cytotoxic T cells and differentiation of CD8 ⁺ cells
IFNs α/β (interferons α/β)	T cells, B cells, monocytes/macrophages, fibroblasts	Antiviral activity, stimulates macrophage activity, modulates MHC class I and II protein expression on various cells, regulates the development of the specific immune response
IFN- γ (interferon- γ)	T cells (T _H 1, T _C), NK cells	Activation of T cells, macrophages, neutrophils, and NK cells; increases class I and II MHC molecules
TNF- α (tumor necrosis factor- α [cachectin])	T cells, macrophages and NK cells	A wide variety of effects due to its ability to mediate expression of genes for growth factors and cytokines, transcription factors, receptors, inflammatory mediators, and acute-phase proteins; plays a role in host resistance to infection by serving as an immunostimulant and mediator of the inflammatory response; cytotoxic for tumor cells
TNF- β (tumor necrosis factor- β [lymphotoxin])	T cells, B cells	Same as TNF- α
G-CSF (granulocyte colony- stimulating factor)	T cells, macrophages, neutrophils	Enhances the differentiation and activation of neutrophils
M-CSF (macrophage colony- stimulating factor)	T cells, neutrophils, macrophages, fibroblasts, endothelial cells	Stimulates various functions of monocytes and macrophages, promotes the growth and development of macrophage colonies from undifferentiated precursors

many other virus-infected cells can synthesize IFN- α and IFN- β . Some of the ways in which interferon renders cells resistant to virus infections are described in **figure 31.18**.

Fever

From a physiological point of view, fever results from disturbances in hypothalamic thermoregulatory activity, leading to an increase of the thermal “set point.” In adult humans **fever** is de-

fined as an oral temperature above 98.6°F (37°C) or a rectal temperature above 99.5°F (37.5°C). The most common cause of a fever is a viral or bacterial infection (or bacterial toxins). In almost every instance there is a specific constituent, the endogenous pyrogen, that directly triggers fever production. Examples of these pyrogens includes interleukin-1, IL-6, and tissue necrosis factor that are produced by host macrophages in response to pathogenic microorganisms. After their release, these pyrogens circulate to the hypothalamus and induce neurons to secrete



prostaglandins. Prostaglandins reset the hypothalamic thermostat at a higher temperature, and temperature-regulating reflex mechanisms then act to bring the core body temperature up to this new setting.

The fever induced by a microorganism augments the host's defenses by three complementary pathways (**figure 31.19**): (a) it stimulates leukocytes so that they can destroy the microorganism, (b) it enhances the specific activity of the immune system, and (c) it enhances microbiostasis (growth inhibition) by decreasing available iron to the microorganism. Evidence suggests that some hosts are able to redistribute the iron during a fever in an attempt to withhold it (**hypoferremia**) from the microorganism. Conversely, the virulence of many microorganisms is enhanced with increased iron availability (**hyperferremia**). Gonococci, for example, spread most often during menstruation, a time in which there is an increased concentration of free iron available to these bacteria.

1. What is a cytokine?
2. Define monokine, lymphokine, interleukin, and colony-stimulating factor.
3. Into what four families are cytokines sometimes placed (table 31.3)?
4. How do cytokines play a role as mediators of natural immunity?
5. How do interferons render cells resistant to viruses?
6. How can a fever be beneficial to a host?

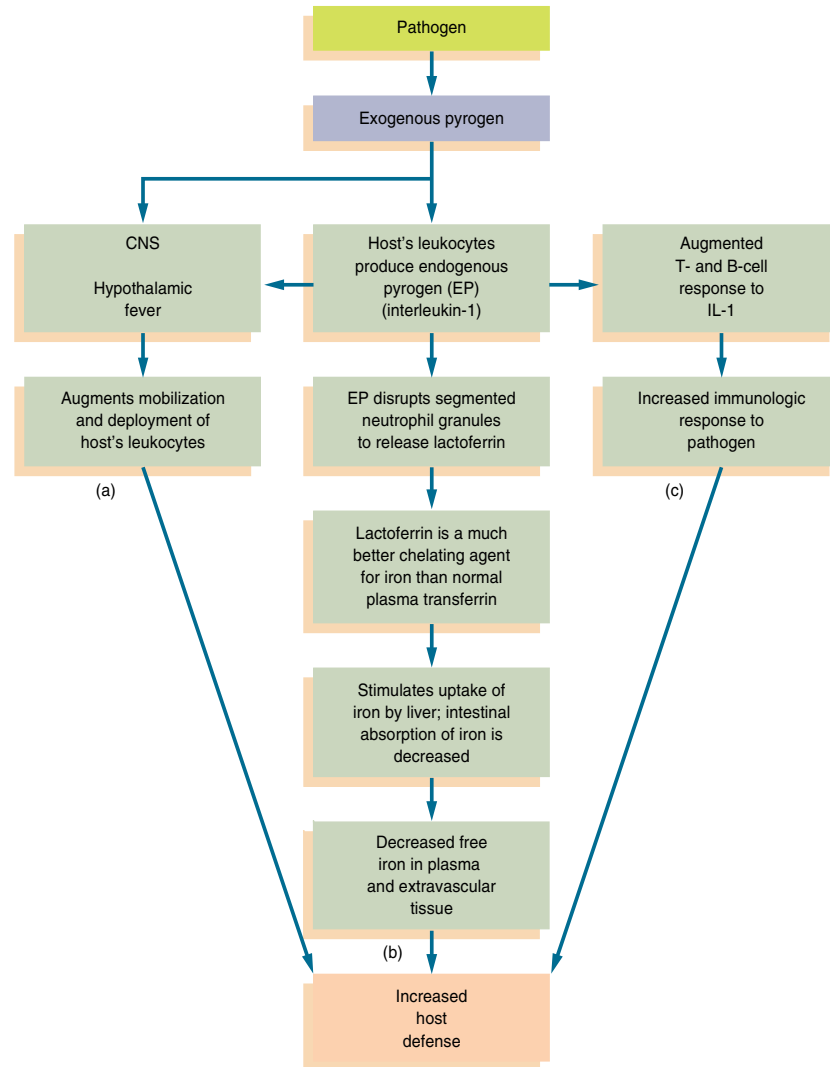
31.10 Natural Killer Cells

Natural killer (NK) cells are a small population of large non-phagocytic granular lymphocytes (figure 31.3). Their major function is to destroy malignant cells and cells infected with microorganisms. They recognize their targets in one of two ways. Like many cells, they possess Fc receptors that bind IgG antibody. These receptors link the NK cells to IgG-coated target cells, which they kill by a process called **antibody-dependent**

Figure 31.18 The Antiviral Action of Interferon. The arrows indicate the sequence of events. **(a)** Interferon (IFN) synthesis and release is often induced by a virus infection or double-stranded RNA (dsRNA). **(b)** Interferon binds to a ganglioside receptor on the plasma membrane of a second cell and triggers the production of enzymes that render the cell resistant to virus infection. The two most important such enzymes are oligo(A) synthetase and a special protein kinase. **(c)** When an interferon-stimulated cell is infected, viral protein synthesis is inhibited by an active endoribonuclease that degrades viral RNA. **(d)** An active protein kinase phosphorylates and inactivates the initiation factor eIF-2 required for viral protein synthesis.

Figure 31.19 Effects of Fever in a Mammalian

Host. (a) One of the first effects of a fever is to accelerate and augment the mobilization and deployment of the host's leukocytes, which in turn isolate and help destroy the pathogen. (b) A fever leads to a reduction of free plasma iron, which limits the growth of those organisms that require a narrow, crucial concentration of iron for their replication and synthesis of toxins. (c) One of the immunologic consequences of fever is the production of endogenous pyrogen (IL-1). IL-1 causes the proliferation, maturation, and activation of lymphocytes (T and B cells), which in turn augment the immunologic response of the host to the pathogen.



cell-mediated cytotoxicity (ADCC) (figure 31.20). The second way NK cells recognize infected cells and cancer cells relies on the killer-activating receptors and killer-inhibitory receptors of the NK cells (**figure 31.21**). The killer-activating receptors recognize a number of different molecules present on the surface of all nucleated cells. The killer inhibitory receptors recognize a surface marker known as the major histocompatibility complex (MHC) class I molecule (*see section 32.4*) that is also present on all nucleated cells and is an indicator of “self.” If the killer-activating receptors are engaged with a ubiquitous surface molecule, a “kill” instruction is issued to the NK cell. Conversely, when this “kill” signal is overridden by an inhibitory signal sent

by the killer inhibitory receptor binding to the MHC class I molecule (which indicates “self”), no killing is done.

Although all nucleated cells normally have the MHC class I molecule on their plasma membrane surface, they can lose this molecule. This loss can occur as a result of a microbial pathogen interfering with the expression mechanism (e.g., after a herpesvirus infection) or from a malignant transformation. Cells that lack this MHC class I molecule are perceived as being abnormal and NK cells do not receive an inhibitory signal. In the absence of an inhibitory signal, NK cells kill these abnormal cells by inserting the pore-forming protein, perforin, into the plasma membrane of the target cell and then injecting it with cytotoxic enzymes called granzymes.

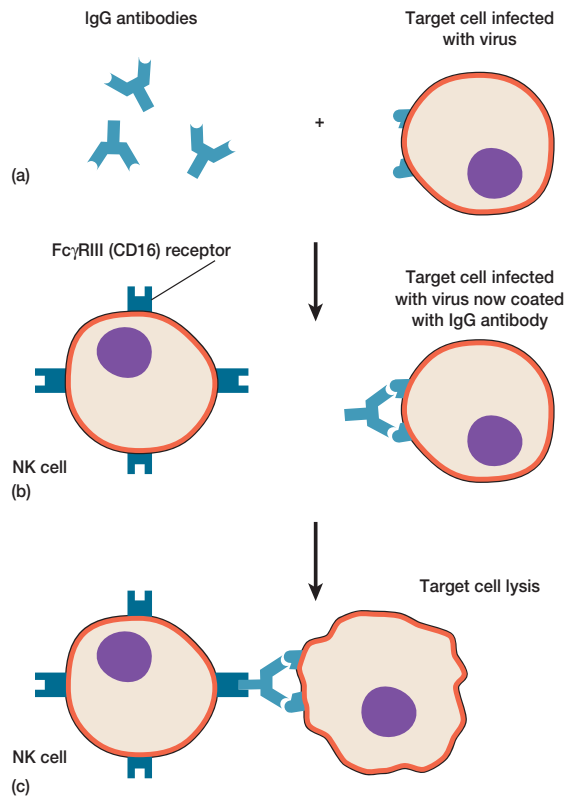


Figure 31.20 Antibody-Dependent Cell-Mediated Cytotoxicity
(a) In this mechanism, IgG antibodies bind to a target cell infected with a virus. (b) NK cells have FcγRIII (CD16) receptors on their surface. (c) When the NK cells encounter infected virus cells coated with IgG antibody, they kill the target cell by releasing cytotoxic mediators and/or membrane damage.

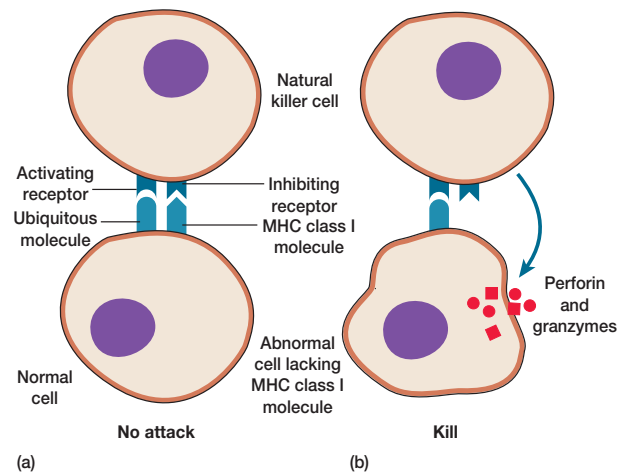


Figure 31.21 The System Used by Natural Killer Cells to Recognize Normal Cells and Abnormal Cells That Lack the Major Histocompatibility Complex Class I Surface Molecule. (a) The killer-activating receptor recognizes a normal ubiquitous molecule on the plasma membrane of a normal cell. Since the killer-inhibitory receptor recognizes the MHC class I molecule, there is no attack. (b) In the absence of the inhibitory signal, the receptor issues an order to the NK cell to attack and kill the abnormal cell. The cytotoxic granules of the NK cell contain perforin and granzymes. With no inhibitory signal, the granules release their contents, killing the abnormal cell.

1. Discuss the role of NK cells in protecting the host.
2. What is the purpose of antibody-dependent cell-mediated cytotoxicity?

Summary

1. Animals and environments that are germfree or have one or more known microorganisms are termed gnotobiotic. Methods are available for rearing gnotobiotic organisms (**figure 31.1**). Gnotobiotic animals and techniques provide good experimental systems with which to investigate the interactions of animals and specific species or microorganisms.
2. Commensal microorganisms living on or in the skin can be characterized as either transients or residents (**figure 31.2**).
3. The normal microbiota of the oral cavity is composed of those microorganisms able to resist mechanical removal.
4. The stomach contains very few microorganisms due to its acidic pH.
5. The distal portion of the small intestine and the entire large intestine have the largest microbial community in the body. Over 400 species have been identified, the vast majority of them anaerobic.
6. The upper genitourinary tract is usually free of microorganisms. In contrast, the adult female genital tract has a complex microbiota.
7. In some instances, after a microorganism contacts or enters a host, a positive mutually beneficial relationship occurs and becomes integral to the health of the host. In other instances, the microorganism may produce disease or even death of the host.
8. Many of the normal host microbiota compete with pathogenic microorganisms.
9. An opportunistic microorganism is generally harmless in its normal environment but becomes pathogenic in a compromised host.
10. There are two fundamentally different types of immune response to invading microorganisms and foreign material. The nonspecific response offers resistance to any microorganism or foreign material. It includes general mechanisms that are a part of the animal's innate structure and function. The nonspecific system has no immunological memory—that is, nonspecific responses occur to the same extent each time. In contrast, the specific response resists a particular foreign agent; moreover, specific immune responses improve on repeated exposure to the agent.

11. The cells responsible for both nonspecific and specific immunity are the white blood cells called leukocytes (**figure 31.3**). Examples include lymphoid cells, mononuclear cells (**figures 31.4 and 31.5**), granulocytes, mast cells, and dendritic cells.
12. Immature undifferentiated lymphocytes generated in the bone marrow mature and become committed to a particular antigenic specificity within the primary lymphoid organs and tissues (**figure 31.6**). In mammals, T cells mature in the thymus and B cells in the bone marrow. The thymus is the primary lymphoid organ and the bone marrow the primary lymphoid tissue.
13. The secondary lymphoid organs and tissues serve as areas where lymphocytes may encounter and bind antigens, whereupon they proliferate and differentiate into fully mature, antigen-specific effector cells. The spleen is a secondary lymphoid organ and the lymph nodes and mucosal-associated tissues (GALT and SALT) are the secondary lymphoid tissues.
14. Many direct factors (age, nutrition) or general barriers contribute in some degree to all host-microbe relationships. At times they favor the establishment of the microorganism; at other times they provide some measure of general defense to the host.
15. Physical and mechanical barriers along with host secretions are the host's first line of defense against pathogens (**figure 31.7**). Examples include the skin (**figure 31.8**) and mucous membranes; the epithelia of the respiratory, gastrointestinal (**figure 31.10**), and genitourinary systems.
16. Mammalian hosts have specific chemical barriers that help combat the continuous onslaught of pathogens. Examples include general chemicals, bacteriocins, and beta-lysins.
17. Inflammation is one of the host's nonspecific defense mechanisms to tissue injury that may be caused by a pathogen (**figures 31.11 and 31.12**). Inflammation can either be acute or chronic.
18. The complement system is composed of a large number of serum proteins (**table 31.1**) that play a major role in the animal's defensive immune response. There are three pathways of complement activation: the classical, alternative, and lectin pathways (**figure 31.13**).
19. Phagocytosis involves the recognition, ingestion, and destruction of pathogens by lysosomal enzymes, superoxide radicals, hydrogen peroxide, defensins, RNIs, and metallic ions (**figure 31.16**; *see also figure 32.31*). Phagocytic cells use two basic mechanisms for the recognition of microorganisms: opsonin-dependent and opsonin-independent.
20. Cytokines (**tables 31.3 and 31.4**) are required for immunoregulation of both the nonspecific and specific immune responses. Cytokines have a broad range of actions on eucaryotic cells (**figure 31.17**).
21. Interferons are a group of cytokines that respond in a defensive way to viral infections, double-stranded RNA, endotoxins, antigenic stimuli, mitogenic agents, and many pathogens capable of intracellular growth (**figure 31.18**).
22. Fever induced by a microorganism augments the host's defenses in three ways (**figure 31.19**): it stimulates leukocytes so they can destroy the invading microorganism; it enhances microbiostasis by decreasing available iron to the microorganism; and it enhances the specific activity of the immune system.
23. Natural killer cells are a small population of large nonphagocytic lymphocytes (**figures 31.20 and 31.21**) that destroy cancer cells and cells infected with microorganisms.

Key Terms

- | | | |
|--|---|--|
| alternative complement pathway 716 | immunity 705 | nonspecific immune response 705 |
| alveolar macrophage 711 | immunology 705 | nonspecific resistance 705 |
| antibody-dependent cell-mediated cytotoxicity (ADCC) 723 | inflammation 712 | normal microbiota 699 |
| bacteriocin 712 | innate or natural immunity 705 | opportunistic microorganism or pathogen 704 |
| basophil 707 | integrin 712 | opsonin-dependent (opsonic) 718 |
| B cell 705 | interdigitating dendritic cell 710 | opsonin-independent (nonopsonic) 718 |
| B lymphocyte 705 | interferon (IFN) 721 | opsonization 718 |
| bronchial-associated lymphoid tissue (BALT) 710 | interleukin 720 | Paneth cell 711 |
| bursa of Fabricius 708 | intraepidermal lymphocyte 710 | pathogen 698 |
| colicin 712 | keratinocyte 709 | pathogenicity 698 |
| colony-stimulating factor (CSF) 720 | lactoferrin 710 | peristalsis 711 |
| comedo 701 | Langerhans cell 709 | phagocytosis 718 |
| complement system 714 | lectin complement pathway 716 | phagolysosome 718 |
| compromised host 704 | leukocyte 705 | phagosome 718 |
| cryptin 711 | lymphocyte 705 | plasma cell 709 |
| cytokine 720 | lymphokine 720 | polymorphonuclear leukocyte (PMN) 707 |
| defensin 720 | lymph node 709 | probiotic 703 |
| dendritic cell 708 | lysozyme 710 | reactive nitrogen intermediate (RNI) 720 |
| ectosymbiosis 701 | M cell 710 | reactive oxygen intermediate (ROI) 718 |
| endosymbiosis 701 | macrophage 705 | respiratory burst 720 |
| eosinophil 707 | mast cell 707 | sebum 701 |
| fever 722 | membrane attack complex 716 | selectin 712 |
| gnotobiotic 698 | monocyte 705 | skin-associated lymphoid tissue (SALT) 709 |
| granuloma 714 | monocyte-macrophage system 705 | specific immune response (acquired or specific immunity) 705 |
| gut-associated lymphoid tissue (GALT) 710 | monokine 720 | spleen 708 |
| hyperferremia 723 | mucociliary blanket 711 | T cell 705 |
| hypoferremia 723 | mucosal-associated lymphoid tissue (MALT) 710 | thymus 708 |
| immune system 705 | natural killer (NK) cell 723 | T lymphocyte 705 |
| | neutrophil 707 | |

Questions for Thought and Review

1. What are some uses of gnotobiotic animals in microbiology?
2. What has the branch of microbiology known as gnotobiotics taught us about the relationship between the human body and the bacteria found on/in it?
3. Describe why the microenvironment of the skin is both favorable to some microorganisms and unfavorable to others. Explain your answer.
4. What is the relationship between a compromised host and an opportunistic microorganism?
5. Compare and contrast the nonspecific and specific immune responses.
6. Describe the different types of human blood cells and give a function for each one.
7. Describe how each of the following systems contributes to the defense of the host: respiratory, gastrointestinal, and genitourinary.
8. How do the SALT and MALT systems operate?
9. How can inflammation be both beneficial and detrimental to a microorganism?
10. What is the relationship between the respiratory burst and phagocytosis?
11. Why is it difficult to group cytokines into specific categories?
12. Why are most fevers caused by either viruses or bacteria?

Critical Thinking Questions

1. Some patients who take antibiotics for acne develop yeast infections. Explain.
2. Some microorganisms are classified as intracellular pathogens. Explain why the nonspecific immune response to these pathogens differs from the response to pathogens that occur on or in the skin.

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CHAPTER 32

Specific Immunity



Nude (athymic) mice have a genetic defect (*nu* mutation) that affects thymus gland development. Thus T cells do not form. They do have a B-cell component and effective humoral immunity. This unique deficiency provides animals in which to study B/T-cell dichotomy and environmental influences on the maturation and differentiation of T cells, as well as many different immune disorders.

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Concepts

1. The major function of the specific immune response in vertebrates is to provide protection (immunity) against harmful microorganisms, cancer cells, and certain macromolecules.
2. Specific immunity has two branches: humoral and cell-mediated. Humoral immunity, or antibody-mediated immunity, involves the production of antibodies by plasma cells derived from B cells. Cell-mediated immunity involves specialized lymphocytes called T cells that act against microorganisms and foreign tissues. They also regulate the activation and proliferation of other immune system cells such as macrophages, B cells, and other T cells.
3. Nonself foreign substances, such as proteins, nucleoproteins, polysaccharides, and some glycolipids to which lymphocytes respond are called antigens.
4. There are five immunoglobulin (Ig) classes based on physicochemical and biological properties. These are IgG, IgM, IgD, IgA, and IgE.
5. Two distinguishing characteristics of immunoglobulins are their diversity and specificity.
6. Various types of antigen-antibody reactions occur in vertebrates that lead to immune product formation. The union of antigen and antibody initiates the participation of other elements that determine the fate of the antigen. For example, the classical complement pathway can be activated, leading to cell lysis or phagocytosis. Other defensive mechanisms include toxin neutralization, opsonization, and immune complex formation.

The remarkable capacity of the immune system to respond to many thousands of different substances with exquisite specificity saves us all from certain death by infection.

—Martin C. Raff

The immune system is responsible for the ability of a host to resist foreign invaders. It includes an array of cells and molecules with specialized roles in defending the host against the continuous onslaught of microbial invaders, and cancer cells. Chapter 31 presents a basic discussion of nonspecific host resistance or innate immunity. Chapter 32 continues this discussion of the immune response by describing the system of specific immunity used to protect the host. Specific immunity can be acquired either naturally or artificially and involves antigens, antibodies, T and B cells, and the classical complement pathway. These aspects of immune system function are presented in this chapter in the continuing discussion of the immune system.

32.1 Overview of Specific Immunity

The specific immune system of vertebrates has three major functions: to recognize anything that is foreign to the body (“non-self”), to respond to this foreign material, and to remember the foreign invader. The recognition response is highly specific. The immune system is able to distinguish one pathogen from another, to identify cancer cells, and to recognize the body’s own “self” proteins and cells as different from “nonself” proteins, cells, tissues, and organs. After recognition of an invader has occurred, the specific immune system responds by recruiting its defensive molecules and cells to attack the invader. This is called an **effector response**. The effector response either eliminates the foreign material or renders it harmless to the host, thus preventing disease. If

the invader is encountered at a later time, the immune system remembers, and mounts a more intense and rapid memory or anamnestic response that eliminates the invader once again and protects the host from disease (p. 743).

Four characteristics distinguish specific immunity from non-specific resistance:

1. **Specificity.** Immunity is directed against a particular pathogen or foreign substance, and the immunity to this pathogen or substance usually does not confer immunity to others.
2. **Memory.** When reexposed to the same pathogen or substance, the body reacts so quickly that there is no noticeable pathogenesis. By contrast, the reaction time for inflammation and other nonspecific defenses is just as long for a later exposure as it was for the initial one.
3. **Diversity.** The system is able to generate an enormous diversity of molecules such as antibodies that recognize billions of different antigens.
4. **Discrimination between self and nonself.** The specific immune system almost always responds only to nonself antigens and thus does not destroy the organism it is defending.

Two branches or arms of specific immunity are recognized (**figure 32.1**): humoral (antibody-mediated) immunity and cellular (cell-mediated) immunity. **Humoral (antibody-mediated) immunity**, named for the fluids or “humors” of the body, is based on the action of soluble proteins called antibodies that occur in the body fluids and on the plasma membranes of B lymphocytes. Circulating antibodies bind to bacteria, toxins, and extracellular viruses, neutralizing them or “tagging or marking” them for destruction by mechanisms described in section 32.6.

Cellular (cell-mediated) immunity is based on the action of specific kinds of T lymphocytes (*see section 31.4 and figure 31.3*) that directly attack cells infected with viruses or parasites, transplanted cells or organs, and cancer cells. T cells can lyse these cells or release chemicals (cytokines) that enhance specific immunity and nonspecific defenses such as phagocytosis and inflammation.

Types of Acquired Immunity

Acquired immunity refers to the type of specific immunity a host develops after exposure to a suitable antigen, or after transfer of antibodies or lymphocytes from an immune donor. Acquired immunity can be obtained by natural or artificial means and actively or passively (**figure 32.2**).

Naturally Acquired Immunity

Naturally acquired active immunity occurs when an individual’s immune system contacts an antigenic stimulus such as an infection. The immune system responds by producing antibodies and activated lymphocytes that inactivate or destroy the antigen. The immunity produced can be either lifelong, as with measles or chickenpox, or last for only a few years, as with tetanus.

Naturally acquired passive immunity involves the transfer of antibodies from one host to another. For example, some of a pregnant woman’s antibodies pass across the placenta to her fetus.

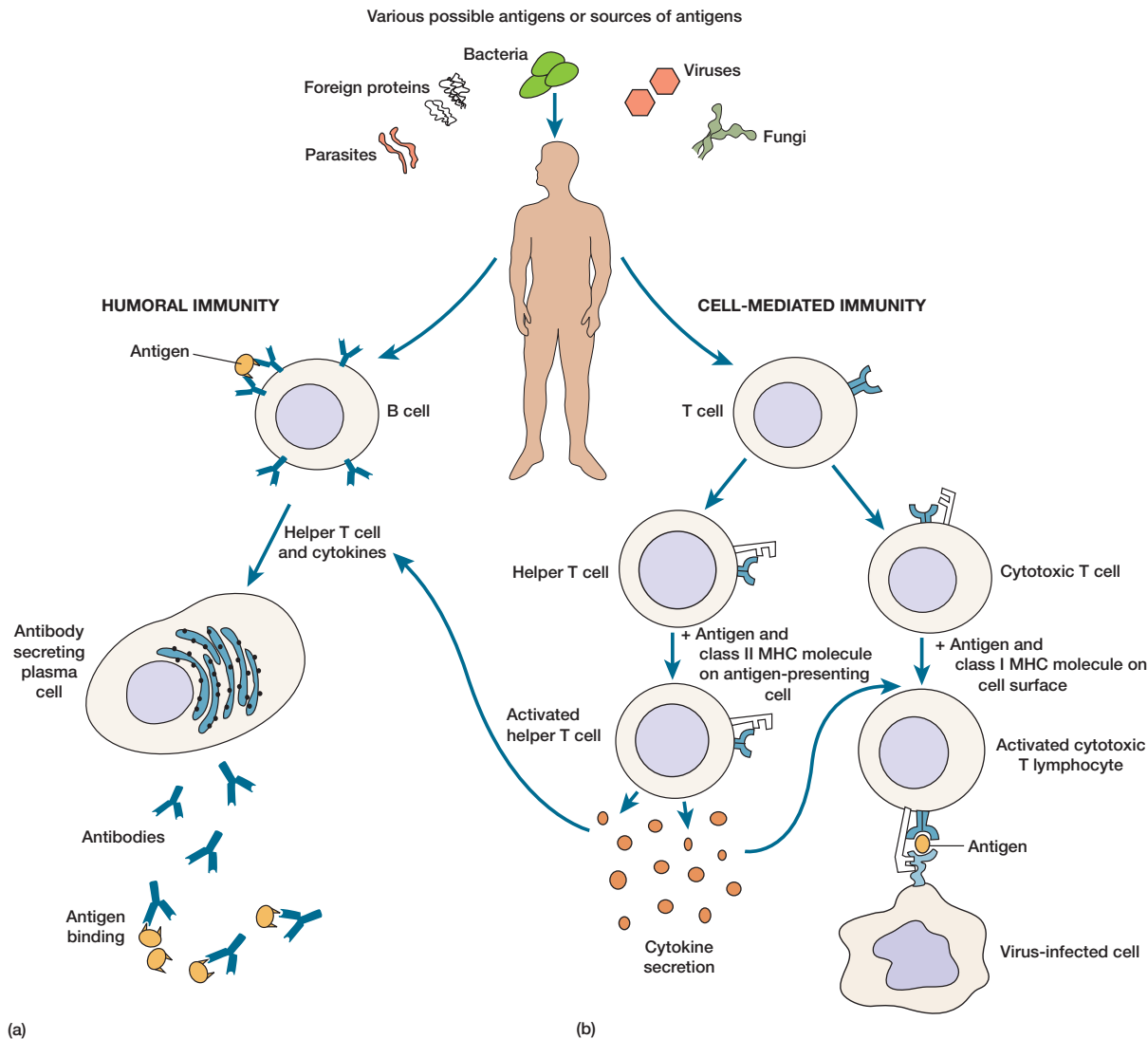


Figure 32.1 The Humoral and Cell-Mediated Branches of Immunity. (a) In the humoral branch, B cells interact with antigen and differentiate into antibody-secreting plasma cells. The antibody binds to the antigen and tags it for destruction by other mechanisms. (b) In the cell-mediated response, subpopulations of T cells are activated by antigen presented in the presence of a major histocompatibility complex (MHC). Activated helper T cells respond by producing a group of specific chemicals called cytokines that facilitate both the humoral and cell-mediated responses. Cytotoxic T cells respond to the antigen by developing into cytotoxic T lymphocytes that kill virus-infected cells and other altered cells. B cells and T cells also differentiate into memory cells (not shown) that are involved in subsequent responses.

If the female is immune to diseases such as polio or diphtheria, this placental transfer also gives the fetus and newborn immunity to these diseases. Certain other antibodies can pass from the female to her offspring in the first secretions (called colostrum) from the mammary glands. These maternal antibodies are essential for providing immunity to the newborn until its own immune system matures. Unfortunately naturally acquired passive immunity generally lasts only a short time (weeks or months at the most).

Artificially Acquired Immunity

Artificially acquired active immunity results when an animal is given an antigen preparation to induce the formation of antibodies and activated lymphocytes. This preparation is called a vaccine and the procedure is vaccination (immunization). A vaccine consists of a preparation of killed microorganisms; living, weakened (attenuated) microorganisms; or inactivated bacterial toxins

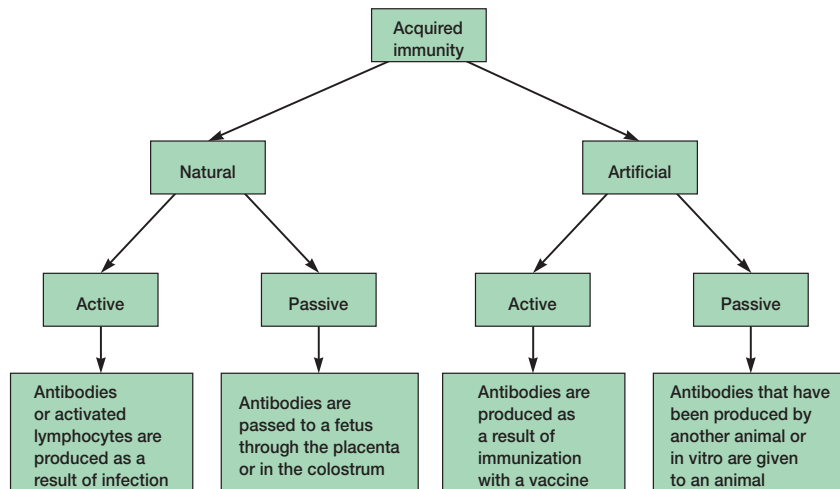


Figure 32.2 Specific Immunity. This illustration can be used as a guide to the text description of acquired immunity. The details of the immune system are astonishingly complex; many events have been omitted so that the overall organization of the immune response can be clearly seen.

(toxoids) that are administered to an animal to induce immunity artificially. Vaccines and immunizations are discussed in detail in section 33.1.

Artificially acquired passive immunity results when antibodies that have been produced either in an animal or by specific methods in vitro are introduced into a host. Although this type of immunity is immediate, it is short lived and lasts only a few weeks to a few months. An example would be botulinum antitoxin produced in a horse and given to a human suffering from botulism food poisoning.

1. The specific immune response can be divided into three related activities. What are these activities?
2. What distinguishes specific immunity from nonspecific resistance?
3. What are the two arms of specific immunity?
4. How does naturally acquired immunity occur? Contrast active and passive immunity.

32.2 Antigens

The immune system distinguishes between “self” and “nonself” through an elaborate recognition process. Prior to birth the body inventories the proteins and various other large molecules present (self) and removes most T cells specific for self-determinants. Subsequently self-substances can be distinguished from nonself substances, and lymphocytes can produce specific immunologic reactions against the latter, leading to their removal.

Substances, such as proteins, nucleoproteins, polysaccharides, and some glycolipids, that elicit an immune response and react with the products of that response are called **antigens** (an-

tibody generator). Most antigens are large, complex molecules with a molecular weight generally greater than about 10,000. The ability of a molecule to function as an antigen depends on its size and structural complexity.

Each antigen can have several **antigenic determinant sites** or **epitopes** (figure 32.3). Epitopes are the regions or sites in the antigen that bind to the antigen-binding site of a specific antibody or with a T cell receptor. Antibodies are formed most readily in response to determinants that project from the foreign molecule or to terminal residues of a specific polymer chain. Chemically, determinants include sugars, organic acids and bases, amino acid side chains, hydrocarbons, and aromatic groups.

The number of antigenic determinant sites on the surface of an antigen is its **valence**. The valence determines the number of antibody molecules that can combine with the antigen at one time. If one determinant site is present, the antigen is monovalent. Most antigens, however, have more than one determinant site or more than one copy of the same epitope, and are termed multivalent. Multivalent antigens generally elicit a stronger immune response than do monovalent antigens.

Haptens

Many small organic molecules are not antigenic by themselves but can become antigenic if they bond to a larger carrier molecule such as a protein (figure 32.4). They cannot stimulate antibody formation or T-cell responses by themselves, but can react with antibodies once formed. Such small molecules are termed **haptens** [Latin *haptein*, to grasp]. When lymphocytes are stimulated by the combined molecule, they can react to either the hapten or the larger carrier molecule. Conjugation of a hapten to a carrier protein makes the hapten immunogenic because the carrier protein can be processed and presented to specific T cells. As a result, both hapten-specific

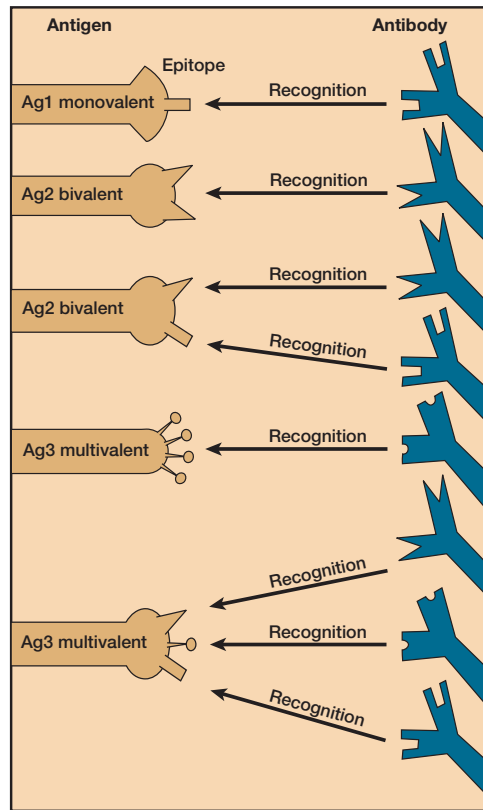


Figure 32.3 The Number of Antigenic Determinant Sites or Epitopes on an Antigen Is Its Valence. Antigen (Ag) molecules each have a set of antigenic determinants or epitopes. There is a single epitope on a monovalent antigen (Ag1). A bivalent antigen (Ag2) may have two of the same epitopes or two different ones, and a multivalent antigen (Ag3) may have many of the same epitopes or three or more different epitopes. In each case, however, a single antibody recognizes only the epitope rather than the whole bivalent or multivalent antigen.

and carrier-specific antibodies can be made. One example of a hapten is penicillin. By itself penicillin is not antigenic. However, when it combines with certain serum proteins of sensitive individuals, the resulting molecule does initiate a severe and sometimes fatal allergic immune reaction. In these instances the hapten is acting as an antigenic determinant on the carrier molecule.

Superantigens

Certain antigens provoke such a drastic immune response that they are termed **superantigens**. Superantigens are bacterial pro-

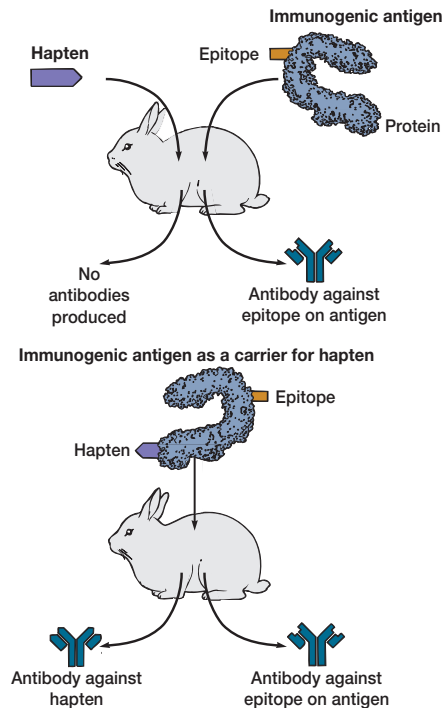


Figure 32.4 Effect of Carrier On Immunogenicity of Hapten.

Haptens, as small molecules, are incapable of inducing an immune response when presented alone. This is markedly different from what occurs when a more complex, immunogenic antigen is presented. In this instance antibody directed against the specific epitope(s) of that antigen can be produced. However, if hapten is linked to a large molecule (which is used as a carrier), the hapten may then be immunogenic. Under these conditions antibody is produced, not only against the epitope of the immunogenic carrier molecule but against the epitope of the hapten as well.

teins. These superantigens nonspecifically stimulate T cells to proliferate by interacting with both class II major histocompatibility complex (section 32.4) products on antigen-presenting cells and the T-cell receptor (**figure 32.5**). Good examples of superantigens are the staphylococcal enterotoxins that cause food poisoning and the toxin that causes toxic shock syndrome. Superantigens cause symptoms by stimulating the release of massive quantities of cytokines from T cells and should be considered possible chronic associates in such diseases as rheumatic fever, arthritis, Kawasaki syndrome, atopic dermatitis, and one type of psoriasis.

Cluster of Differentiation Molecules (CDs)

The involvement of lymphocytes with the immune response and various diseases can be assessed by enumeration of specific cells bearing particular membrane proteins called **cluster of differentiation molecules (CDs)**. CDs are functional cell surface

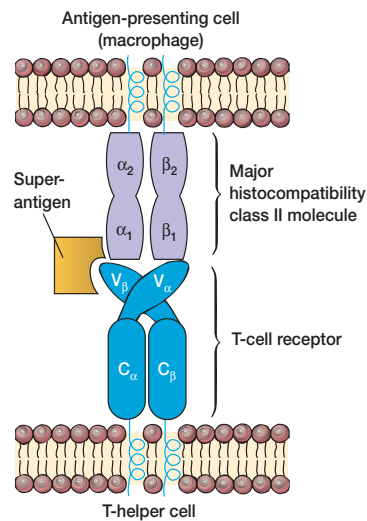


Figure 32.5 Superantigens. The most prevalent antigen receptor of T cells is the T-cell receptor. Superantigens stimulate T cells bearing a particular V_β element of the T-cell receptor, regardless of the nature of the V_α receptor element. They bind directly to the outer surface of the MHC class II molecule without being processed by the antigen-presenting cell. Thus a superantigen can activate many more T cells than a normal antigen does.

proteins or receptors that can be measured in situ and from peripheral blood, biopsy samples, or other body fluids. They often are used as a nomenclature system to differentiate between leukocyte subpopulations. To date, over 200 CDs have been characterized. **Table 32.1** summarizes some of the functions of several CDs.

CDs have both biological and diagnostic significance. In normal individuals the concentration of these molecules in serum is very low. When one's immune system is activated in response to disease, the concentration of these molecules usually rises and fluctuates. Monitoring CD levels may help in the management of the disease. For example, circulating levels of CD54 are directly related to the progress and prognosis of the skin cancer metastatic melanoma. CD23 is a B-cell growth factor, and elevated levels are associated with chronic lymphocytic leukemia. CD35 is a potent agent for the suppression of complement-dependent tissue injury in autoimmune inflammatory disease. Elevated levels of CD8 have been found in childhood lymphoid malignancies and in HIV-infected individuals. It has been established that the CD4 molecule is a cell surface receptor for HIV-1 (see figure 38.9), and considerable research is currently being done on the role of both cell-bound and soluble CD4 in AIDS.

1. Define and give several examples of an antigen.
2. What is an antigenic-determinant site or epitope?
3. Describe a hapten.
4. What are some examples of the biological significance of cell-associated differentiation antigens?
5. Distinguish between "self" and "nonself" substances.

Table 32.1 Some Functional Examples of Cluster of Differentiation Molecules (CDs)

CD Molecule	Function
CD2	The CD2 molecule is a glycoprotein present on T cells, thymocytes and NK cells. It functions as an intercellular adhesion molecule and binds to the human leukocyte function-associated antigen-3 or LFA-3 (CD58). CD2 may be involved in the attachment of T_H cells to antigen-presenting cells and cytotoxic T cells to target cells, and in other cellular interactions. It also is a signal-transducing molecule and may help stimulate T cells to proliferate and secrete cytokines.
CD4	The CD4 molecule is a glycoprotein found primarily on T cells associated with helper/inducer regulatory functions and to a lesser degree on monocytes and macrophages. It is a cell adhesion molecule with great affinity for class II MHC.
CD8	The CD8 molecule is a polypeptide found on the surface of cytotoxic T cells. It is a molecule that participates in the interaction of class I MHC found on target cells.
CD23	CD23 is a protein found on the surface of IgM-bearing B cells, eosinophils, macrophages, and platelets. Soluble CD23 is a B-cell growth factor and upregulates IgE synthesis in conjunction with interleukin-4 released from T cells.
CD35	CD35 is a glycoprotein present on erythrocytes, PMNs, monocytes, all B and some T cells, mast cells, and in soluble form in the plasma. It is the receptor for complement C3b/C4b. The functions of CD35 vary according to the cell type possessing the receptor. For example, CD35 on erythrocytes binds immune complexes bearing C3b or C4b and promotes their clearance from the circulation. On B cells, CD35 augments the maturation of these cells into antibody-secreting cells. It also aids leukocytes in binding and phagocytosing C3b- or C4b-coated objects.
CD54	CD54, also known as intercellular adhesion molecule-1 (ICAM-1), is a protein that is an important early marker of immune activation and response. It is a ligand for the lymphocyte function-associated antigen-1 (LFA-1) found on lymphocytes, monocytes, and neutrophils. The CD54/LFA-1 interaction is important in T-cell-antigen-specific responses and lymphocyte emigration into inflammatory sites. CD54 also is found on mucosal epithelial cells and erythrocytes. Recent evidence suggests CD54 is the cellular receptor for the human rhinovirus and <i>Plasmodium falciparum</i> -infected erythrocytes.

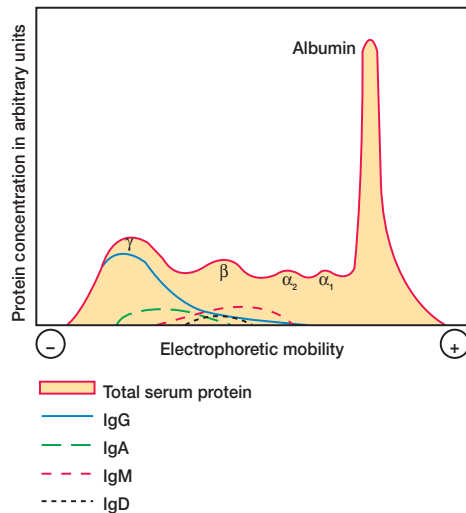


Figure 32.6 Electrophoresis of Human Serum. Schematic representation of electrophoretic results illustrating the distribution of serum proteins and four major classes of immunoglobulins.

32.3 Antibodies

An **antibody** or **immunoglobulin (Ig)** is a glycoprotein that is made in response to an antigen, and can recognize and bind to the antigen that caused its production. Antibodies are present in the blood serum, tissue fluids, and mucosal surfaces of vertebrate animals. Serum glycoproteins can be separated according to their charge by movement through a gel in an electric field and classified as albumin, alpha-1 globulin, alpha-2 globulin, beta globulin, and gamma globulin (**figure 32.6**). The gamma globulin band is relatively wide because it contains a heterogeneous class of immunoglobulins—namely, IgG, IgA, IgM, and IgD. These differ from each other in molecular size, structure, charge, amino acid composition, and carbohydrate content. Most antibodies in serum are in the IgG class. A fifth immunoglobulin class, the IgE class, has a similar mobility to IgD but cannot be represented quantitatively because of its low concentration in serum.

Immunoglobulin Structure

Antibodies have more than one antigen-combining site. For example, most human antibodies have two combining sites or are bivalent (**figure 32.3**). Some bivalent antibody molecules can combine to form multimeric antibodies that have up to 10 combining sites.

All immunoglobulin molecules have a basic structure composed of four polypeptide chains (**figure 32.7a,b**) connected to each other by disulfide bonds. Each light chain usually consists of about 220 amino acids and has a mass of approximately 25 kDa. Each heavy chain consists of about 440 amino acids and has a mass of about 50 to 70 kDa. The heavy chains are structurally distinct for each immunoglobulin class or subclass. Both light

and heavy chains contain two different regions. **Constant regions** (C_L and C_H) have amino acid sequences that do not vary significantly between antibodies of the same class. The **variable regions** (V_L and V_H) from different antibodies do have different sequences (**figure 32.7a**). It is the variable regions (V_L , V_H) that when folded together form the antigen binding sites.

The four chains are arranged in the form of a flexible Y with a hinge region. This hinge region allows the antibody molecule to assume a T shape. The stalk of the Y is termed the **crystallizable fragment (Fc)** and contains the site at which the antibody molecule can bind to a cell. The top of the Y consists of two **antigen-binding fragments (Fab)** that bind with compatible epitopes (or antigenic determinant sites). The Fc fragments are composed only of constant regions, whereas the Fab fragments have both constant and variable regions. Both the heavy and light chains contain several homologous units of about 100 to 110 amino acids. Within each unit, called a **domain**, disulfide bonds form a loop of approximately 60 amino acids (**figure 32.7c**). Interchain disulfide bonds also link heavy and light chains together.

More specifically the light chain may be either of two distinct forms called kappa (κ) and lambda (λ). These can be distinguished by the amino acid sequence of the carboxyl portion of the chain (**figure 32.8a**). In human immunoglobulins, the carboxyl-terminal portion of all κ chains is identical; thus this region is termed the constant (C_L) domain. With respect to the lambda chains, there are four very similar sequences that define the subtypes λ_1 , λ_2 , λ_3 , and λ_4 with their corresponding constant regions $C_{\lambda 1}$, $C_{\lambda 2}$, $C_{\lambda 3}$, and $C_{\lambda 4}$. Within the light chain variable domain are hypervariable regions or complementarity-determining regions (CDRs) that differ in amino acid sequence more frequently than the rest of the variable domain.

In the heavy chain the NH_2 -terminal domain has a pattern of variability similar to that of the V_κ and V_λ domains and is termed the V_H domain. However, notice in **figure 32.8b** that this domain contains four hypervariable regions. The other domains of the heavy chains are termed constant domains and are numbered C_{H1} , C_{H2} , C_{H3} , and sometimes C_{H4} , starting with the domain next to the variable domain (**figure 32.7c**). The constant domains of the heavy chain form the constant (C_H) region. The amino acid sequence of this region determines the classes of heavy chains. In humans there are five classes of heavy chains designated by lowercase Greek letters: gamma (γ), alpha (α), mu (μ), delta (δ), and epsilon (ϵ). The properties of these heavy chains determine, respectively, the five immunoglobulin classes—IgG, IgA, IgM, IgD, and IgE. Each immunoglobulin class differs in its general properties, half-life, distribution in the body, and interaction with other components of the host's defensive systems.

Within two of the major human immunoglobulin classes, there are variants. These variations can be classified as (1) **isotypes**, referring to the variations in the heavy chain constant regions associated with the different classes that are normally present in all individuals (**figure 32.9a**); (2) **allotypes**, the genetically controlled allelic forms of immunoglobulin molecules that are not present in all individuals (**figure 32.9b**); and (3) **idiotypes**,

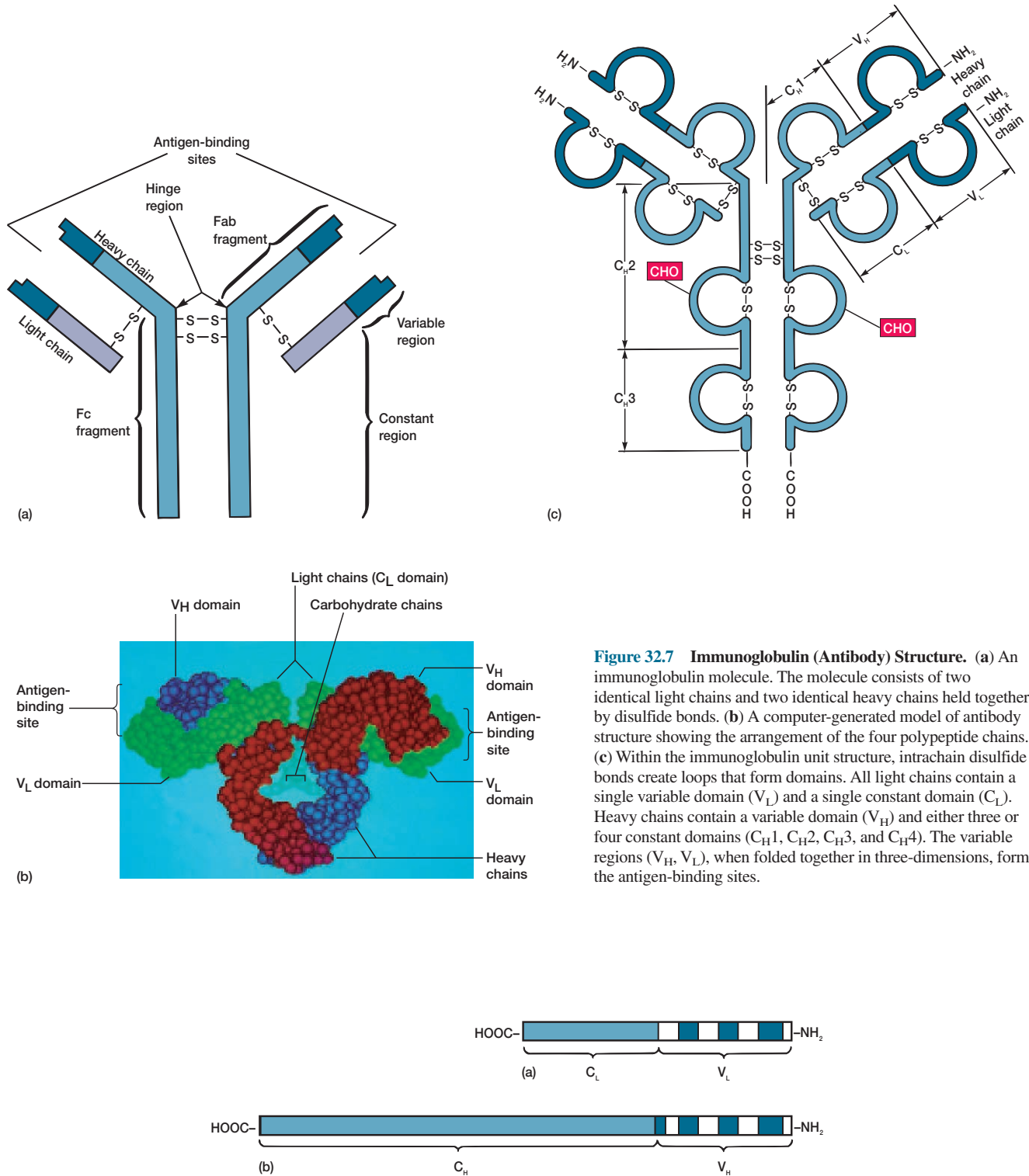


Figure 32.8 Constant and Variable Domains. Location of constant (C) and variable (V) domains within (a) light chains and (b) heavy chains. The dark blue bands represent hypervariable regions or complementarity-determining regions within the variable domains.

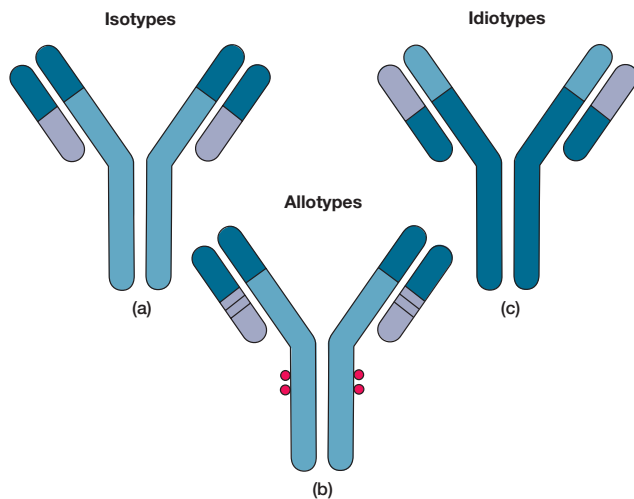


Figure 32.9 Variants of Immunoglobulins. (a) Isotypes represent variants present in serum of a normal individual. (b) Allotypes represent alternative forms coded for by different alleles and so are not present in all individuals of a population. (c) Idiotypes are individually specific to each immunoglobulin molecule.

referring to individual specific immunoglobulin molecules that differ in the hypervariable region of the Fab portion (figure 32.9c). The many variations of immunoglobulin structure reflect the diversity of antibodies generated by the immune response.

1. Name the two types of antibody light chains.
2. What is the function of the Fc region of an antibody? The Fab region?
3. What is the variable region of an antibody? The hypervariable or complementarity-determining region? The constant region?
4. What determines the class of heavy chain within an antibody?
5. Name the five immunoglobulin classes.
6. Distinguish between isotype, allotype, and idio type.

Immunoglobulin Function

Each end of the immunoglobulin molecule has a different role. The Fab region is concerned with binding to antigen, whereas the Fc region mediates binding to host tissue, various cells of the immune system, some phagocytic cells, or the first component of the complement system. The binding of an antibody with an antigen usually does not cause destruction of the antigen or of the microorganism, cell, or agent to which it is attached. Rather the antibody serves to mark and identify the target for immunologic attack and to activate nonspecific immune responses that can destroy the target. For example, bacteria that are covered with antibodies are better targets for phagocytosis by neutrophils and macrophages. The alteration of the surface of bacteria, viruses, and other particles so that they can be more readily phagocytized

is termed opsonization (*see section 31.8*). Immune destruction also is promoted by antibody-induced activation of the classical complement system as discussed in section 32.7.

Immunoglobulin Classes

IgG is the major immunoglobulin in human serum, accounting for 80% of the immunoglobulin pool (**figure 32.10a**). IgG is present in blood plasma and tissue fluids. The IgG class acts against bacteria and viruses by opsonizing the invaders and neutralizing toxins. It is also one of the two immunoglobulin classes that activate complement by the classical pathway (section 32.7; *see also section 31.7*). IgG is the only immunoglobulin molecule able to cross the placenta and provide natural immunity in utero and to the neonate at birth.

There are four human IgG subclasses (IgG1, IgG2, IgG3, and IgG4) that vary chemically in their chain composition and the number and arrangement of interchain disulfide bonds (**figure 32.10b**). About 65% of the total serum IgG is IgG1, and 23% is IgG2. Differences in biological function have been noted in these subclasses. For example, IgG2 antibodies are opsonic and develop in response to antitoxins. Anti-Rh antibodies are of the IgG1 or IgG3 subclass. IgG1 and IgG3, upon recognition of their specific antigens, bind to receptors expressed on monocytes and macrophages and make them better phagocytes. These receptors are termed Fc receptors. The IgG4 antibodies function as skin-sensitizing immunoglobulins (*see section 33.2*).

IgM accounts for about 10% of the immunoglobulin pool. It is usually a polymer of five monomeric units (pentamer), each composed of two heavy chains and two light chains (**figure 32.11**). The monomers are arranged in a pinwheel array with the Fc ends in the center, held together by disulfide bonds and a special **J (joining) chain**. IgM is the first immunoglobulin made during B-cell maturation and is expressed as membrane-bound antibody on B cells. IgM is secreted into serum during a primary antibody response (**figure 32.19b**). Since IgM is so large, it does not leave the bloodstream or cross the placenta. IgM agglutinates bacteria, activates complement by the classical pathway, and enhances the ingestion of pathogens by phagocytic cells.

Although most IgM appears to be pentameric, around 5% or less of human serum IgM exists in a hexameric form. This molecule contains six monomeric units but seems to lack a J chain. Hexameric IgM activates complement up to twentyfold more effectively than does the normal pentameric form. It has been suggested that bacterial cell wall antigens such as gram-negative lipopolysaccharides may directly stimulate B cells to form hexameric IgM without a J chain. If this is the case, the immunoglobulins formed during primary immune responses (**figure 32.19a**) are less homogeneous than previously thought.

IgA accounts for about 15% of the immunoglobulin pool. Some IgA is present in the serum as a monomer of two heavy and two light chains. Most IgA, however, occurs in mucous secretions as a polymerized dimer held together by a J chain (**figure 32.12**). IgA has special features that are associated with secretory mucosal surfaces. IgA, when transported from the mucosal-associated

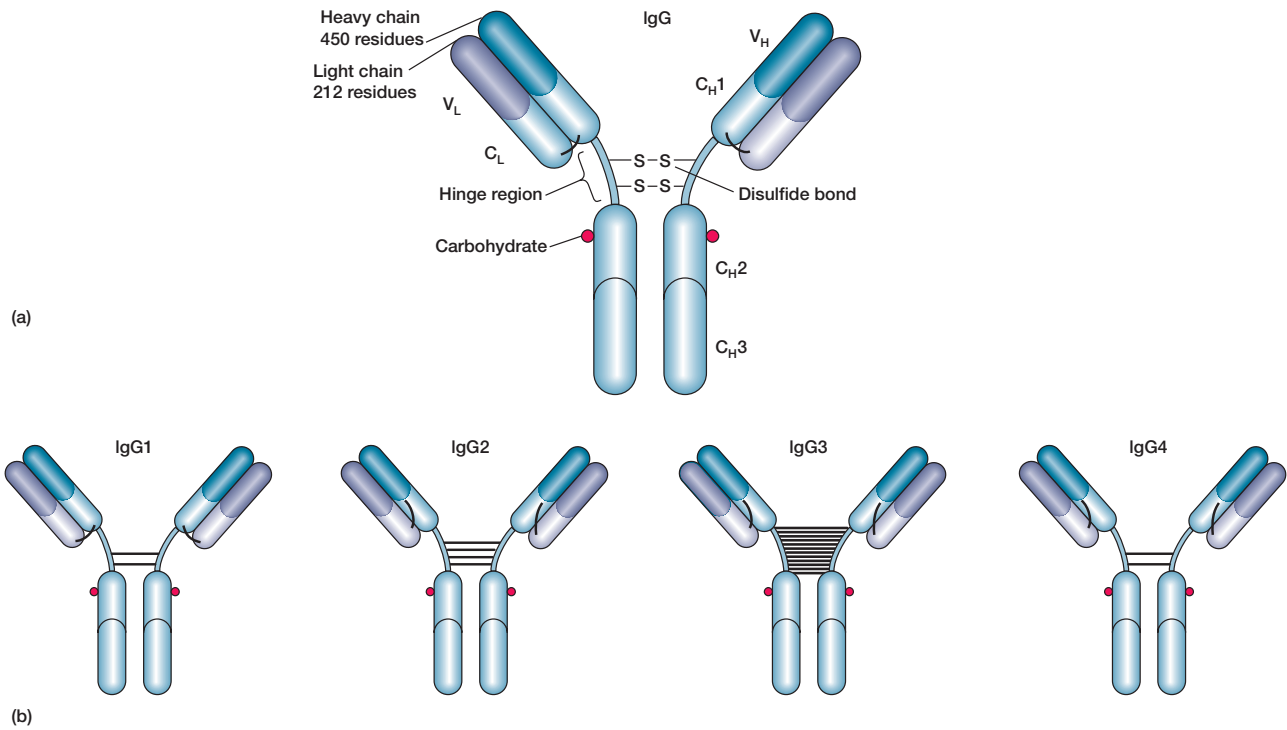


Figure 32.10 Immunoglobulin G. (a) The basic structure of human IgG. (b) The structure of the four human IgG subclasses. Note the arrangement and numbers of disulfide bonds (shown as thin black lines).

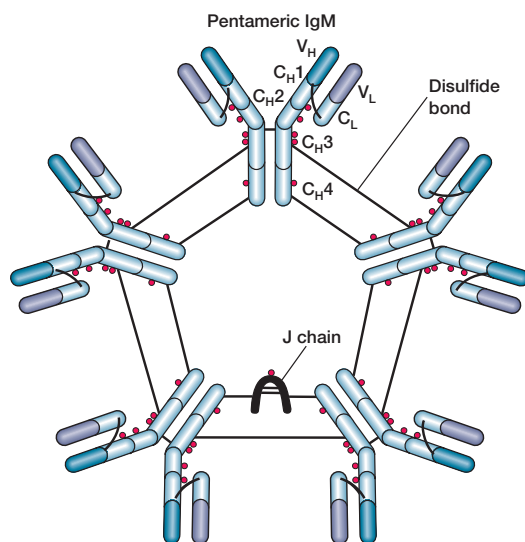


Figure 32.11 Immunoglobulin M. The pentameric structure of human IgM. The disulfide bonds linking peptide chains are shown in black; carbohydrate side chains are in red. Note that 10 antigen-binding sites are present.

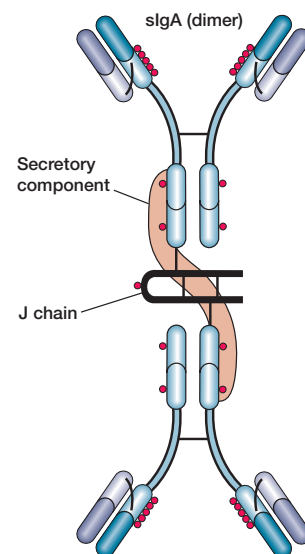


Figure 32.12 Immunoglobulin A. The dimeric structure of human secretory IgA. Notice the secretory component (tan) wound around the IgA dimer and attached to the constant domain of each IgA monomer.

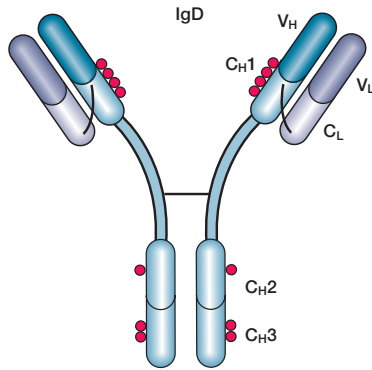


Figure 32.13 Immunoglobulin D. The structure of human IgD. The disulfide bonds linking protein chains are shown in black; carbohydrate side chains are in red.

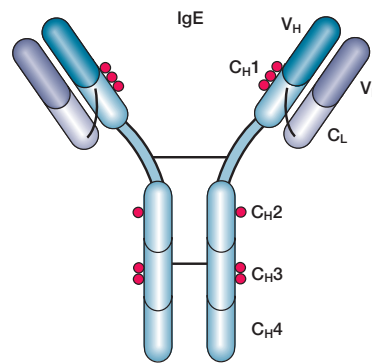


Figure 32.14 Immunoglobulin E. The structure of human IgE.

lymphoid tissue to mucosal surfaces, acquires a protein termed the secretory component. **Secretory IgA (sIgA)**, as the modified molecule is now called, is the primary immunoglobulin of mucosal-associated lymphoid tissue (*see figure 31.10*). Secretory IgA is also found in saliva, tears, and breast milk. In these fluids and related body areas, sIgA plays a major role in protecting surface tissues against infectious microorganisms by the formation of an immune barrier. For example, in breast milk sIgA helps protect nursing newborns. In the intestine, sIgA attaches to viruses, bacteria, and protozoan parasites such as *Entamoeba histolytica*. This prevents pathogen adherence to mucosal surfaces and invasion of host tissues, a phenomenon known as immune exclusion. In addition, sIgA binds to antigens within the mucosal layer of the small intestine; subsequently the antigen-sIgA complexes are excreted through the adjacent epithelium into the gut lumen. This rids the body of locally formed immune complexes and decreases their access to the circulatory system. Secretory IgA also plays a role in the alternative complement pathway (*see section 31.7*).

IgD is an immunoglobulin found in trace amounts in the blood serum. It has a monomer structure (**figure 32.13**) similar to IgG. IgD antibodies do not fix complement and cannot cross the placenta, but they are abundant in combination with IgM on the surface of B cells and bind antigens, thus signaling the B cell to start antibody production.

IgE (**figure 32.14**) makes up only a small percent of the total immunoglobulin pool. The classic skin-sensitizing and anaphylactic antibodies belong to this class. IgE molecules have four constant region domains ($C_{\epsilon 1}$, $C_{\epsilon 2}$, $C_{\epsilon 3}$, and $C_{\epsilon 4}$) on their heavy chains. The Fc portion of the $C_{\epsilon 4}$ chain can bind to special Fc_{ϵ} receptors on mast cells and basophils. When two IgE molecules on the surface of these cells are cross-linked by binding to the same antigen, the cells degranulate. This degranulation releases histamine and other pharmacological mediators of anaphylaxis. It also stimulates eosinophilia and gut hypermotility (increased rate of movement of the intestinal contents) that aid in the elimination of helminthic parasites. Thus, though IgE is present in small amounts, this class of antibodies has very potent biological capabilities. [Anaphylaxis \(pp. 768–69\)](#)

Table 32.2 summarizes some of the more important physicochemical properties of the human immunoglobulin classes.

1. What does antigen binding accomplish in a host?
2. Give the major functions of each immunoglobulin class.
3. Why is the structure of IgG considered the model for all five immunoglobulin classes?
4. Which immunoglobulin can cross the placenta?
5. Which immunoglobulin is most prevalent in the immunoglobulin pool? The least prevalent?

Diversity of Antibodies

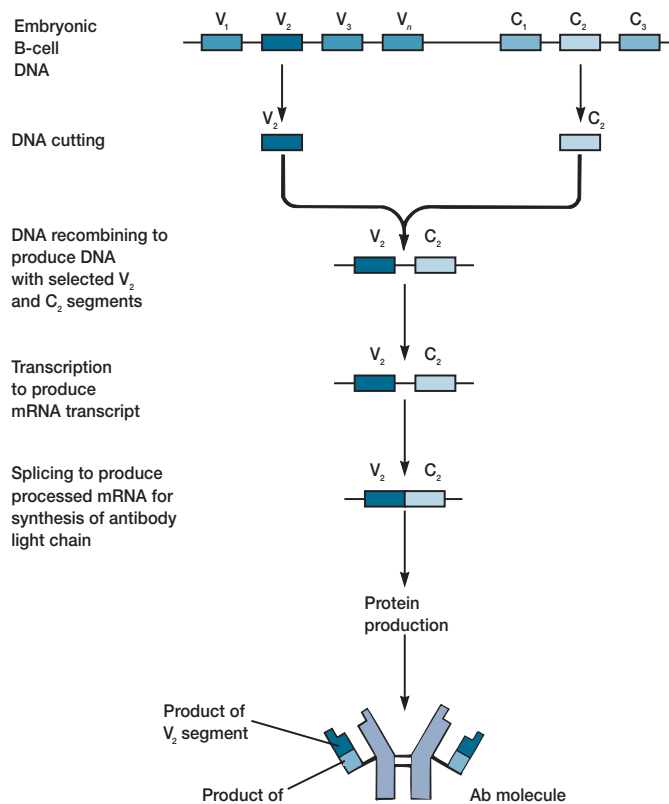
One unique property of antibodies is their remarkable diversity. According to current estimates each human can synthesize more than 10^{11} (100 billion) different kinds of antibodies. How is this diversity generated? The answer is threefold: rearrangement of antibody gene segments, somatic mutations, and generation of different codons during antibody gene splicing.

Immunoglobulin genes are split or interrupted genes with many gene segments (*see section 12.1*). Embryonic B cells contain a small number of gene segments, close together on the same chromosome, that determine the constant (C) region of the light chains (**figure 32.15**). Separated from them, but on the same chromosome, is a larger cluster of segments that determines the variable (V) region of the light chains. During B-cell differentiation, one segment for the constant region is joined by a process of recombination mediated by specific proteins to one segment for the variable region. This splicing produces a complete light-chain antibody gene. A similar splicing mechanism also occurs to join the constant and variable segments of the heavy chains.

Because the light-chain genes actually consist of three parts, and the heavy-chain genes consist of four, the formation of a finished antibody molecule is slightly more complicated than previ-

Table 32.2 Physicochemical Properties of Human Immunoglobulin Classes

Property	Immunoglobulin Classes				
	IgG ^a	IgM	IgA ^b	IgD	IgE
Heavy chain	γ	μ	α	δ	ϵ
Mean serum concentration (mg/ml)	9	1.5	3.0	0.03	0.00005
Percent of total antibody	80–85	5–10	5–15	<1	0.002–0.05
Valency	2	5(10)	(2–4)	2	2
Mass of heavy chain (kDa)	51	65	56	70	72
Mass of entire molecule (kDa)	146	970	160 ^c	184	188
Placental transfer	+	–	–	–	–
Half-life in serum (days) ^d	23	5	6	3	2
Complement activation					
Classical pathway	++	+++	–	–	–
Alternative pathway	–	–	+	–	–
Induces mast cell degranulation	–	–	–	–	+
Major characteristics	Most abundant Ig in body fluids; neutralizes toxins, opsonizes bacteria, activates complement, maternal antibody	First to appear after antigen stimulation; very effective agglutinator; expressed as membrane-bound antibody on B Cells	Secretory antibody; protects external surfaces	Present on B-cell surface; B-cell recognition of antigen	Anaphylactic-mediating antibody; resistance to helminths
% carbohydrate	3	7–10	7	12	11

^aProperties of IgG subclass 1.^bProperties of IgA subclass 1.^csIgA = 360 – 400 kDa^dTime required for half of the antibodies to disappear.

ously outlined (figure 32.16). The germ line DNA for the light-chain gene contains multiple coding sequences called V and J (joining) regions. During the differentiation of a B cell, a deletion (which is variable in length) occurs that joins one V gene segment with one J segment. This DNA joining process is termed combinatorial joining because it can create many combinations of the V and J regions. When the light-chain gene is transcribed, transcription continues through the DNA region that encodes for the constant portion of the gene. RNA splicing subsequently joins the VJ and C regions, creating mRNA.

Combinatorial joining in the formation of a heavy-chain gene occurs by means of DNA splicing of the heavy-chain counterparts of V and J along with a third set of D (diversity) sequences (figure 32.17a). Initially, all heavy chains have the μ type of constant region. This corresponds to antibody class IgM (figure 32.17b). Another DNA splice joins the VDJ region with a different constant region that can subsequently change the class of antibody produced by the B cell (figure 32.17c).

Figure 32.15 Gene Shuffling and Antibody Diversity. Antibody diversity is partly the result of the shuffling of gene sequences that code for both heavy and light chains. This drawing shows the shuffling, cutting, and splicing process used to produce an assembled light chain of the antibody molecule.

Figure 32.16 Light Chain Production in a Mouse. One V segment is randomly joined with one J-C region by deletion of the intervening DNA. The remaining J segments are eliminated from the RNA transcript during RNA processing. An intron is a segment of DNA occurring between expressed regions of genes.

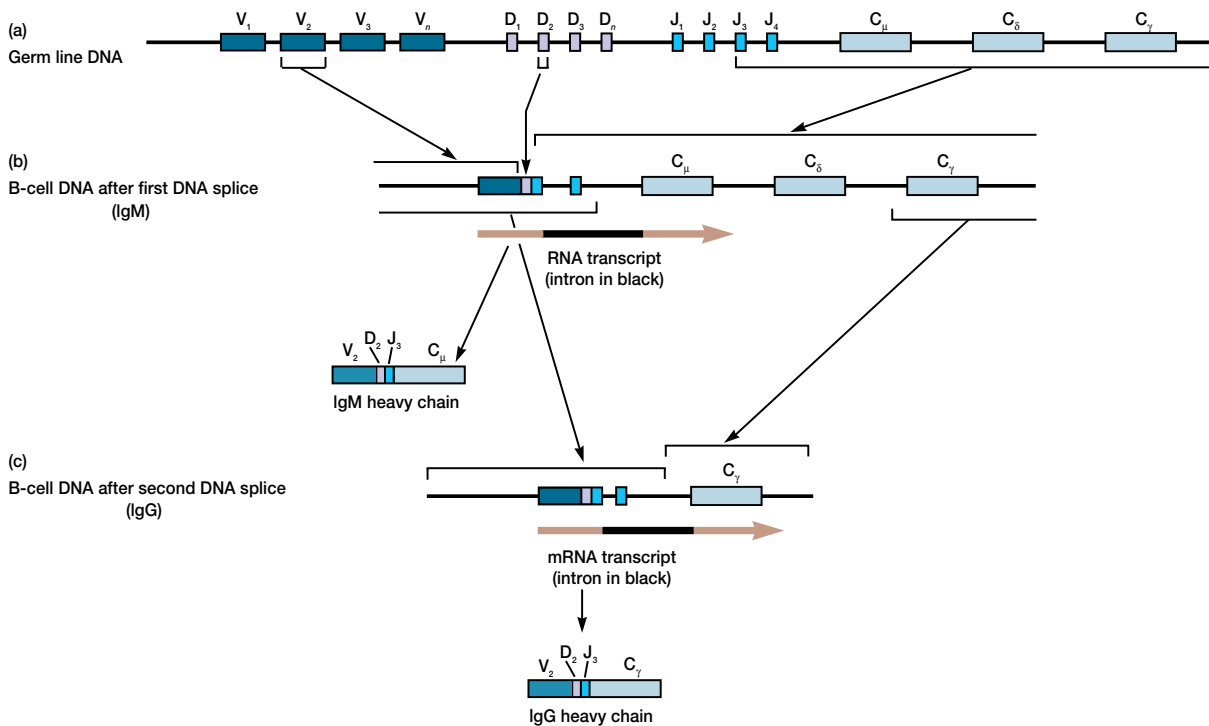
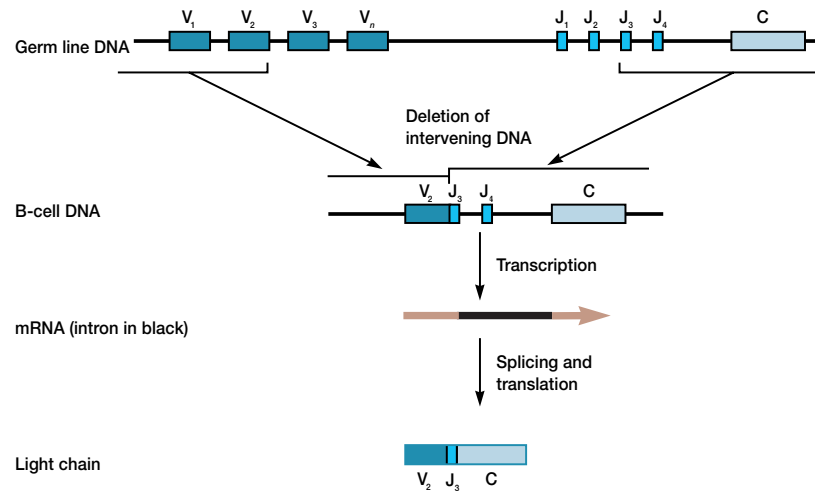


Figure 32.17 The Formation of a Gene for the Heavy Chain of an Antibody Molecule. See text for further details.

The amount of antibody diversity in the mouse that can be generated by combinatorial joining is shown in **table 32.3**. In this animal the κ light chains can be formed from any combination of about 250 to 350 V $_{\kappa}$ and 4 J $_{\kappa}$ regions giving a maximum of approximately 1,400 different κ chains. The λ chains have their own V $_{\lambda}$ and J $_{\lambda}$ regions but

smaller in number than their κ counterparts (6 different λ chains). The heavy chains have approximately 250 to 1,000 V $_{H}$, 10 to 30 D, and 4 J $_{H}$ regions, giving a maximum of 120,000 different combinations. Because any light chain can combine with any heavy chain, there will be a maximum of 2×10^8 possible κ chain antibody types.

Table 32.3 Number of Antibodies Possible through the Combinatorial Joining of Mouse Germ Line Genes^a

λ light chains	V regions = 2 J regions = 3 Combinations = $2 \times 3 = 6$
κ light chains	V _κ regions = 250–350 J _κ regions = 4 Combinations = $250 \times 4 = 1,000$ $= 350 \times 4 = 1,400$
Heavy chains	V _H = 250–1,000 D = 10–30 J _H = 4 Combinations = $250 \times 10 \times 4 = 10,000$ $= 1,000 \times 30 \times 4 = 120,000$
Diversity of antibodies	κ-containing: $1,000 \times 10,000 = 10^7$ $1,400 \times 120,000 = 2 \times 10^8$ λ-containing: $6 \times 10,000 = 6 \times 10^4$ $6 \times 120,000 = 7 \times 10^5$

^aApproximate values.

The value of 2×10^8 different antibodies is actually an underestimate because antibody diversity is further augmented by two processes:

1. The V regions of germ line DNA are susceptible to a high rate of somatic mutation during B-cell development in response to an antigen challenge. These mutations allow B-cell clones to produce antibodies with somewhat different polypeptide sequences.
2. The junction for either VJ or VDJ splicing in combinatorial joining can occur between different nucleotides and thus generate different codons in the spliced gene. For example, one VJ splicing event can join the V sequence CCTCCC with the J sequence TGGTGG in two ways:



which codes for the amino acids proline and tryptophan; and



which codes for proline and arginine. Thus the same VJ joining could produce polypeptides differing in a single amino acid.

1. How many chromosomes encode for antibody production in humans?
2. What is the name of each gene segment set that encodes for the different regions of antibody chains?
3. Describe what is meant by combinatorial joining of VJ and C regions of a chromosome.
4. In addition to combinatorial joining, what other two processes play a role in antibody diversity?

Specificity of Antibodies

As noted previously, combinatorial joinings, somatic mutations, and variations in the splicing process generate the great variety of antibodies produced by mature B cells. From a large, diverse B-cell pool, specific cells are stimulated by antigens to reproduce and form a B-cell clone whose cells contain the same genetic information. This is known as the **clonal selection theory**, a hypothesis to explain immunologic specificity and memory.

The existence of a B-cell **clone** (a population of cells derived asexually from a single parent) that can respond to an antigen by producing the correct antibody is the first tenet of this theory. The lymphoid system is thus considered to contain many B-cell clones, each clone able to recognize a specific antigen. The antigen selects the appropriate B cell to form a clone (hence the phrase “clonal selection”), and the cells with other antigen specificities are unaffected.

According to the second tenet, each B-cell is genetically programmed to respond to its own distinctive antigen before the antigen is introduced. The particular antibody an individual B cell can produce is integrated into the plasma membrane of that B cell and acts as a specific surface receptor for the corresponding antigen molecule. The reaction of the antibody and antigen initiates the differentiation and multiplication of the B cell to form two different cell populations: plasma cells and **memory B cells** (figure 32.18).

Plasma cells are literally protein factories that produce about 2,000 antibodies per second in their brief five- to seven-day life span. Memory B cells can initiate the antibody-mediated immune response upon detecting the particular antigen specific for their antibody molecules. These memory cells circulate more actively from blood to lymph and live much longer (years or even decades) than plasma cells. Memory cells are responsible for the immune system’s rapid secondary antibody response (figure 32.19) to the same antigen. Finally, memory B cells and plasma cells are usually not produced unless the B cell has interacted with, and received cytokine signals from, activated helper T cells (figure 32.1).

Sources of Antibodies

The need for pure homogeneous antibodies has increased dramatically in recent years. Currently antibodies are produced either naturally by immunization or artificially through hybridoma formation.

Immunization

Specific antibodies can be produced naturally by the immunization of domestic animals or human volunteers. Whatever the source, purified antigen is injected into the host. The host’s immune system recognizes and responds to the antigen, and its B cells proliferate and differentiate to produce specific antibodies. To promote the efficiency of antigen stimulation of antibody production, the antigen may be mixed with an **adjuvant** [Latin *adjuvans*, aiding], which enhances the rate and quantity of antibody produced. Following repeated antigen injections at regular intervals, blood is withdrawn from the host and allowed to clot. The fluid that remains after the

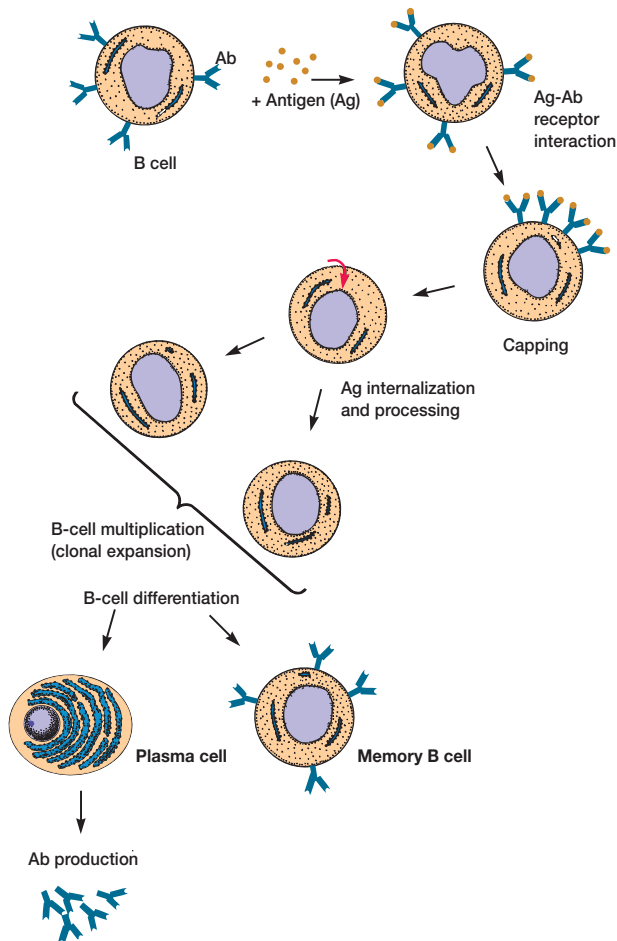


Figure 32.18 Clonal Selection. The immune system can respond specifically to a myriad of possible antigens, whether they are individual molecules or are attached to pathogens and abnormal cells such as cancer cells. B cells or B lymphocytes constantly roam the body, particularly the blood and lymphoid tissues. Each B cell synthesizes only one of the millions of possible antibodies and displays this antibody on its surface. When the antigen meets a B cell having a surface antibody of the proper specificity (top left), it complexes with the antibody and capping occurs. (Capping is the regional aggregation of antibodies on the surface of the cell following Ag-Ab interaction.) The antigen is then internalized; the B cell swells and begins to divide rapidly, producing a B-cell clone. The activated B-cell clone differentiates into plasma cells and memory cells. Plasma cells form the specific antibody that immediately attacks the antigen that provoked its formation. Memory B cells persist in the body and boost the immune system's readiness to eliminate the same antigen if it presents itself in the future. Most antibody responses also require signals from T helper cells (not shown).

blood clots is the **serum** (Latin "whey"). Because this serum has been obtained from an immunized host and contains the desired antibodies, it is called **antiserum**.

Although antiserum is a major and convenient source of antibodies, its usefulness is limited in three ways:

1. Antibodies obtained by this method are polyclonal; they are produced by several B-cell clones and have different

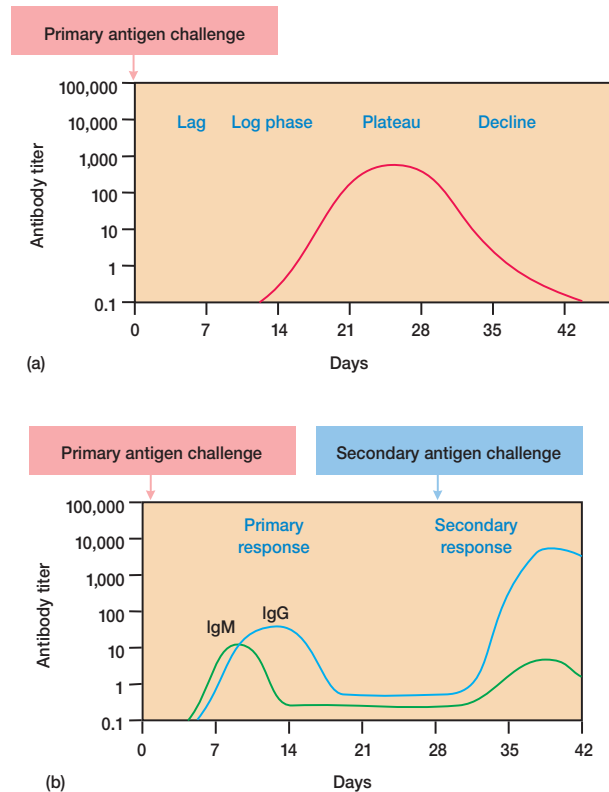


Figure 32.19 Primary and Secondary Antibody Responses. (a) The four phases of a primary antibody response. (b) Comparison of primary and secondary antibody responses. Note that the secondary response to the antigen is both faster and greater than the primary response. T-cell-mediated responses exhibit similar immunologic memory.

specificities. This decreases their sensitivity to particular antigens and results in some degree of cross-reaction with closely related antigen molecules.

2. Second or repeated injections of antiserum from one species into another can cause serious allergic or hypersensitivity reactions.
3. Antiserum contains a mixture of antibodies (figure 32.6), not all of which are of interest in a given immunization.

The Primary Antibody Response

During immunization procedures (and also in naturally acquired immunity), there is an initial lag phase of several days following a primary challenge with an antigen. During the lag phase no antibody can be detected in the blood (**figure 32.19a**). The antibody **titer**, which is the reciprocal of the highest dilution of an antiserum that gives a positive reaction in the test being used, rises logarithmically to a plateau during the second or log phase. In the plateau phase the antibody titer stabilizes. This is followed by a decline phase, during which antibodies are naturally metabolized or bound to the antigen and cleared from the circulation. During the primary antibody response, IgM appears first, then IgG (figure

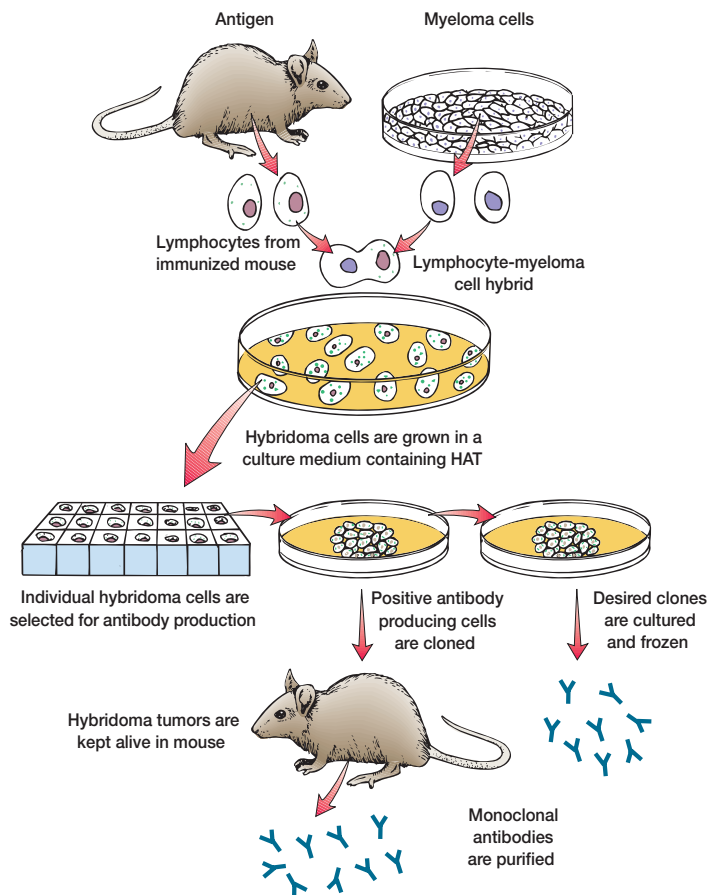


Figure 32.20 Technique for the Production of Monoclonal Antibodies. Lymphocytes are fused with special mutant myeloma cells, yielding hybridomas. Each of them secretes a single, “monoclonal” antibody. Once the hybridoma secreting the desired antigen is identified, it is cloned to generate many antibody-secreting cells that yield the huge quantity of a single antibody needed in medicine or science. Some hybridoma cells may be stored frozen and later cloned for antibody production or kept alive in laboratory animals.

32.19b). The affinity of the antibodies for the antigen’s determinants is low to moderate during this primary antibody response.

The Secondary Antibody Response

The primary antibody response primes the immune system so that it possesses specific immunologic memory through its clones of memory B cells. Upon secondary antigen challenge (figure 32.19b), the B cells mount a heightened secondary or **anamnestic** [Greek *anamnesis*, remembrance] **response** to the same antigen. Compared to the primary antibody response, the secondary antibody response has a shorter lag phase, a more rapid log phase, persists for a longer plateau period, attains a higher IgG titer, and produces antibodies with a higher affinity for the antigen (affinity maturation through somatic hypermutation).

Hybridomas

The limitations of antiserum as a source of antibodies have been overcome with the development of hybridoma techniques to manipulate and culture various mammalian cells that synthesize antibodies in vitro. Each cell and its progeny normally produce a **monoclonal antibody (MAb)** of a single specificity.

The methodology of one of these techniques is illustrated in **figure 32.20**. Animals (usually mice or rats) are immunized with anti-

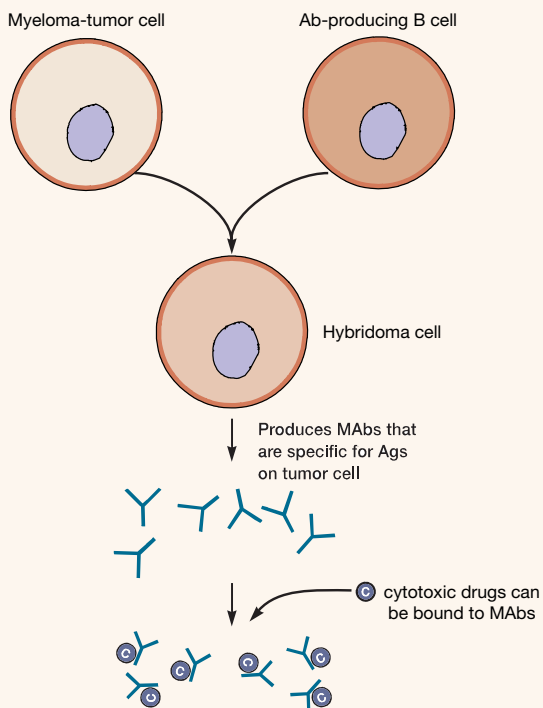
gens as discussed previously. Once the animals are producing a large quantity of antibodies, their spleens are removed and antibody-producing B and plasma cells (lymphocytes) isolated. These lymphocytes are then fused with myeloma cells by the addition of polyethylene glycol, which promotes membrane fusion. **Myeloma cells** are cancerous plasma cells that can readily be cultivated; mutant myeloma cells incapable of producing immunoglobulins are used. These fused cells, derived from lymphocytes and myeloma cells, are called **hybridomas** (they are hybrids of the two cells).

The fusion mixture is then transferred to a culture medium containing a combination of hypoxanthine, aminopterin, and thymidine (HAT). Aminopterin is a poison that blocks a specific metabolic pathway in cells. Myeloma cells lack an enzyme that allows their growth in the presence of aminopterin. However, the pathway is bypassed in lymphoid cells provided with the intermediate metabolites hypoxanthine and thymidine. As a result the hybridomas grow in the HAT medium but the myeloma cells die because they have a metabolic defect and cannot employ the bypass or salvage pathway.

When the culture is initially established using the HAT medium, it contains lymphocytes, myeloma cells, and hybridomas. The unfused lymphoid cells die naturally in culture within a week or two, and the myeloma cells die in the HAT as just described. In contrast, the fused cells survive because they have the immortality of the myeloma and the metabolic bypass of the lymphoid cells. Hybridomas that have the antibody-producing

Box 32.1

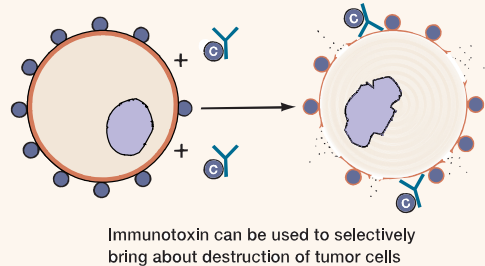
Immunotoxins



One result of hybridoma research is the production of **immunotoxins** (see **Box figure**). Immunotoxins are monoclonal antibodies that have been attached to a specific toxin or toxic agent (antibody + toxin = immunotoxin). Immunotoxins kill target cells and no others, because the antibody binds specifically to plasma membrane surface antigens found only on the target cells. This approach is being used to treat certain types of cancer.

In this procedure cancer cells from a person are injected into mice or rats to stimulate the production of specific antibodies against their plasma membrane antigens. Monoclonal antibodies are produced using hybridomas, purified, and attached to an agent toxic to the cancer cells.

When the immunotoxin is given to a cancer patient, it circulates through the body and binds only to the cancer cells that have the appropriate surface antigens. After binding to the surface, the immunotoxin is taken into cancer cells by receptor-mediated endocytosis, and released inside. The immunotoxin then interferes with the metabolism of the target cells and kills them. Although this procedure is still experimental, it holds great promise in the treatment of certain types of cancer.



capacity of the original lymphoid cells are randomly placed in culture wells. The wells are individually tested for production of the desired antibody, and, if positive, the cells within the well provide clones of immortal cells, all producing the same monoclonal antibody.

Monoclonal antibodies currently have many applications. For example, they are routinely used in the typing of tissue, in the identification and epidemiological study of infectious microorganisms, in the identification of tumor and other surface antigens, in the classification of leukemias, and in the identification of functional populations of different types of T cells. Anticipated future uses include (1) passive immunizations against infectious agents and toxic drugs, (2) tissue and organ graft protection,

(3) stimulation of tumor rejection and elimination, (4) manipulation of the immune response, (5) preparation of more specific and sensitive diagnostic procedures, and (6) delivery of antitumor agents (immunotoxins) to tumor cells (**Box 32.1**).

1. Describe the major two tenets of the clonal selection theory.
2. What two cell populations are produced by a B cell after it initially reacts with an antigen?
3. What is the function of an adjuvant?
4. Describe the difference between serum and antiserum.
5. What is a hybridoma? How is it made?

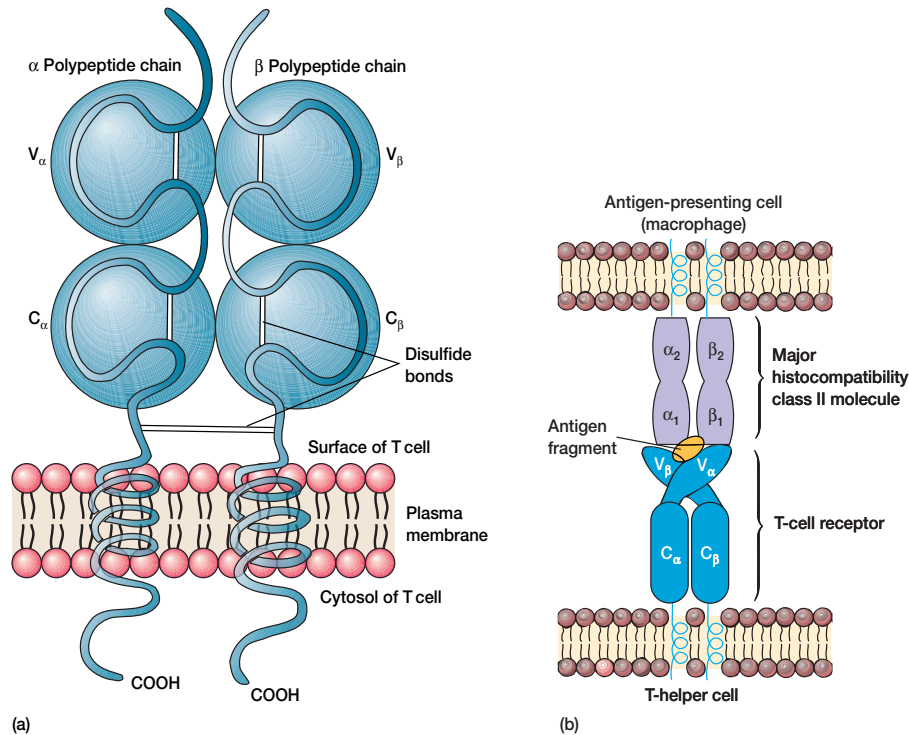


Figure 32.21 The Role of the T-cell Receptor Protein in T helper Cell Activation. (a) A schematic illustration of the proposed overall structure of the antigen receptor site on a T-cell plasma membrane. (b) An antigen-presenting cell begins the activation process by displaying an antigen fragment (e.g., peptide) on its surface as part of a complex with the histocompatibility molecules. A T-helper cell is activated after the variable region of its receptor (designated V_α and V_β) reacts with the antigen fragment and MHC molecule on the presenting cell surface.

32.4 T-Cell Biology

T cells are major players in the cell-mediated immune response (figure 32.1), and also have a major role in B cell activation. They are immunologically specific, can carry a vast repertoire of immunologic memory, and can function in a variety of regulatory and effector ways.

T-Cell Receptors

T cells have specific **T-cell receptors (TCRs)** for antigens on their plasma membrane surface. The receptor site is composed of two parts, an alpha polypeptide chain and a beta polypeptide chain (figure 32.21a). Each chain is stabilized by disulfide bonds. The receptor is anchored in the plasma membrane and parts of the α and β chains extend into the cytoplasm. The recognition sites of the T-cell receptor extend out from the membrane and have a terminal variable section complementary to antigen fragments. T cells respond to antigen fragments exposed on the surfaces of **antigen-presenting cells (APCs)**, most of which are macrophages, dendritic cells, and B cells. These cells scavenge foreign materials (antigens or pathogens) from the blood and tissues and digest them. Inside the cells small pieces of the antigens bind to a special class of proteins called major histocompatibility complex (MHC) class II molecules (section 32.5). The MHC-antigen complex is then transported to the surface of the presenting cell and displayed to passing lymphocytes known as T-helper cells (figure 32.21b). Some knowledge of these histocompatibility molecules is thus re-

quired before T-cell-macrophage interactions and T-cell biology can be understood.

Major Histocompatibility Complex (MHC)

A woman dies because her body has rejected a kidney transplant; a man is crippled with rheumatoid arthritis; an African child goes into a coma that is brought on by cerebral malaria; and another child dies because of an immunodeficiency. Although these four clinical situations are diverse, they all have one thing in common: the cause of all of them involves the human version of the major histocompatibility complex. Malfunction of this system, which is at the basis of these situations, has such wide-ranging effects not only because of the system's role in the specific immune response, but also because of its genetic complexity.

The **major histocompatibility complex (MHC)** is a collection of genes on chromosome 6 in humans and chromosome 17 in mice. This term is derived from the Greek word for tissue [*histo*] and the ability to get along [compatibility]. The MHC is called the **human leukocyte antigen (HLA) complex** in humans and **H-2 complex** in mice. Almost all human tissue cells contain MHC molecules on their plasma membranes. MHC molecules can be divided into three classes: class I molecules are found on almost all types of nucleated body cells; class II molecules appear only on those leukocytes involved in T helper cell-related immune responses (macrophages, antigen-presenting cells [dendritic cells], and B cells); and class III molecules include various secreted proteins that have immune functions (complement components C2,

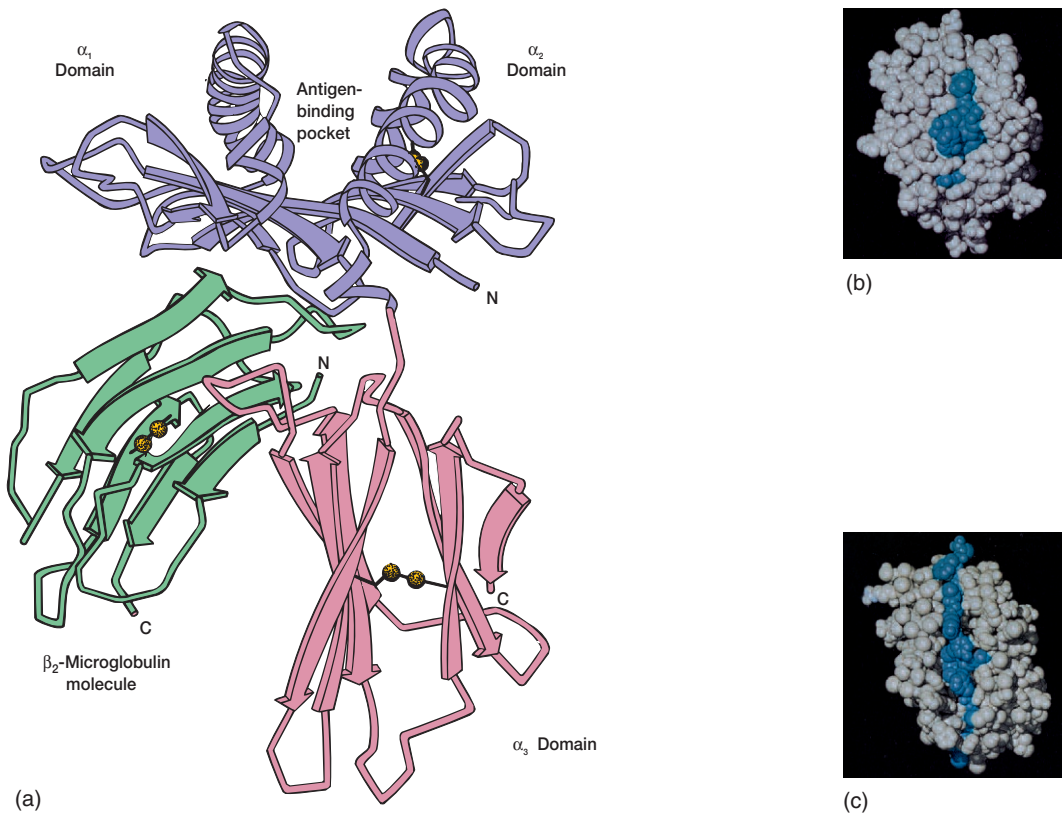


Figure 32.22 Class I Major Histocompatibility Molecule Structure. (a) This side view of the molecule shows the arrangement of its three domains (α_1 , α_2 , α_3) and the β_2 -microglobulin molecule. (b) Class I MHC molecules (space-filling model) can hold only short peptides (shown in blue) because the binding site is closed off. (c) Since the binding site is open on both ends, class II MHC molecules (space-filling model) can bind to peptides (shown in blue) of different lengths.

C4a, and factor B), two steroid 21-hydroxylase enzymes (21-OHA and 21-OHB), the inflammatory cytokines, tumor necrosis factors α and β , and two heat-shock proteins. Unlike class I and II MHC molecules, the class III molecules are not membrane proteins, are not related to class I or II molecules, and have no role in antigen presentation. Thus they will not be discussed further.

Class I MHC molecules (figure 32.22a) consist of a complex of two protein chains, one with a mass of 45 kDa (the heavy chain) and the other with a mass of 12 kDa (β_2 -microglobulin). The two chains contain four regions. The outer segment of the heavy chain can be divided into three functional domains, designated α_1 , α_2 , and α_3 . The β_2 -microglobulin (β_2m) protein and α_3 segment of the heavy chain are noncovalently associated with one another and are close to the plasma membrane. A small segment of the heavy chain is

attached to the membrane by a short amino acid sequence that extends into the cell interior, but the rest of the protein protrudes to the outside. The α_1 and α_2 domains lie to the outside and form the antigen-binding pocket.

Class II MHC molecules are also transmembrane proteins consisting of α and β chains of mass 34 kDa and 28 kDa, respectively. Both chains are folded to give two domains.

Although MHC class I and class II molecules are structurally distinct, both fold into very similar shapes (figure 32.22b,c). Each MHC molecule has a deep groove into which a short peptide can bind. Because this peptide is not part of the MHC molecule, it can vary from one MHC molecule to the next. On healthy cells all of these peptides come from self-proteins. The presence of foreign peptides (antigen fragments) in the MHC groove alerts the immune system and activates T cells, which in turn activate macrophages.

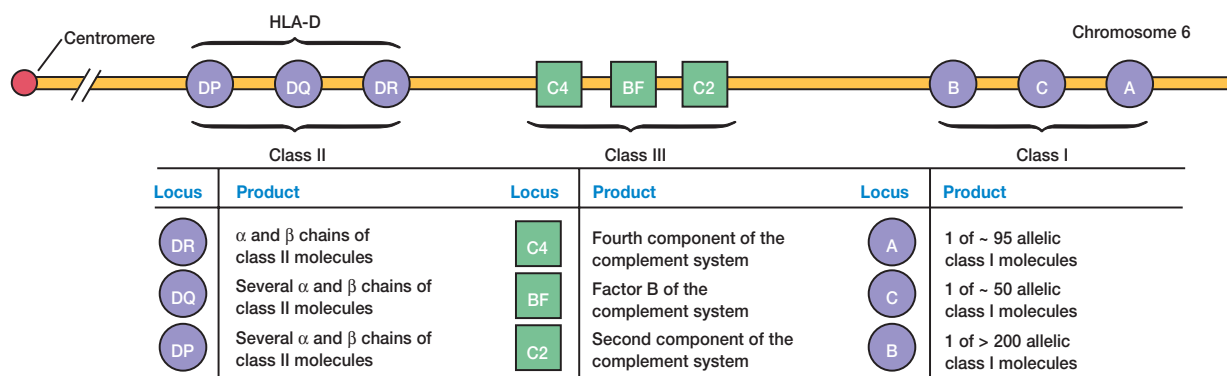


Figure 32.23 Major Histocompatibility Complex. The MHC region of human chromosome 6 and the gene products associated with each locus.

Class I and class II molecules present peptides that arise in different places within cells as the result of a process known as antigen processing. Class I molecules bind to peptides that originate in the cytoplasm (e.g., antigens from replicating viruses). Endogenous antigenic proteins are digested inside the cell as part of the natural process by which a cell continually renews its protein contents. The short peptide fragments that result from this process are pumped by a specific transporter protein from the cytoplasm into the endoplasmic reticulum. Within the endoplasmic reticulum the class I MHC α chain is synthesized and associates with β_2 -microglobulin. This dimer appears to bind peptide as soon as it enters the endoplasmic reticulum. The MHC molecule and peptide are then carried to, and anchored in the plasma membrane. If the peptide is foreign (e.g., short pieces of viral protein), a passing $CD8^+$ T cell (cytotoxic T lymphocyte) whose T cell receptor is specific for the antigenic peptide will bind to the peptide-MHC complex and ultimately kill the infected cell.

Class II MHC molecules bind to fragments that arise from exogenous antigens. This pathway would function with bacteria and viruses that have been taken up endocytotically. The antigen-presenting cell takes in the antigen or pathogen by receptor-mediated endocytosis or phagocytosis and produces antigen fragments by digestion in the phagolysosome. Fragments then combine with preformed class II MHC molecules and are delivered to the cell surface. It is here that the peptide is recognized by $CD4^+$ T-helper cells. Unlike $CD8^+$ T cells, $CD4^+$ T cells do not directly kill target cells. Instead they respond in two distinct ways. One is to enlarge and divide, thereby increasing the number of $CD4^+$ cells that can react to the antigen. The second response is to secrete cytokines (e.g., interleukin-2) that either directly inhibit the pathogen that produced the antigen or recruit and stimulate other cells to join in the immune response.

As previously noted, MHC molecules are coded by a group of genes termed the major histocompatibility complex (figure

32.23). The MHC of humans is located on chromosome 6 and contains three major classes of genes. Many forms of MHC genes exist because the presence of multiple alleles that have arisen by high gene mutation rates, gene recombination, and other mechanisms. Each individual has two sets of these genes, one from each parent, and both are expressed (i.e., they are codominant). Thus a person expresses many different MHC products. The MHC proteins will differ between individuals; the closer two people are related, the more similar are their MHC molecules.

Class I molecules are made by all cells of the body except red blood cells and comprise MHC types A, B, and C. Class I molecules serve to identify almost all cells of the body as “self.” They also stimulate antibody production when introduced into a host with different class I molecules. This is the basis for MHC typing when a patient is being prepared for an organ transplant (**Box 32.2**).

Class II molecules comprise the D group of the MHC and are produced only by activated macrophages, dendritic cells, mature B cells, some T cells, and certain cells of other tissues. Class II molecules are required for T-cell communication with macrophages and B cells. As discussed in more detail later, part of the T-cell receptor must recognize a peptide and a class II molecule on the antigen-presenting cell before the T cell can secrete cytokines necessary for the immune response.

The class III genes encode the second component of complement (C2), factor B, two forms of the fourth component of complement (C4a and C4b), tumor necrosis factor, heat shock proteins, and other proteins. C2, C4a, and C4b participate in the classical pathway and factor B in the alternative pathway (see figure 31.13).

Classes I and II of the MHC are involved in not only the immunologic recognition of microorganisms but also an individual’s susceptibility to particular noninfectious diseases. For example, there is evidence of MHC-linked determinants in tuberculoid leprosy, paralytic poliomyelitis, multiple sclerosis,

Box 32.2

Donor Selection for Tissue or Organ Transplants

The likelihood of tissue or organ transplant acceptance can be increased by using persons who have a great degree of genetic similarity. The greater the similarity, the more likely the persons will have similar MHC complexes. The reason for this is that 77 histocompatibility alleles are determined by just four human histocompatibility genes—A, B, C, and D (which includes DQ, DR, and DP). These genes are transmitted from parents to offspring according to Mendelian genetics. For example, two siblings (brother and sister) may have approximately a 25% chance of being B identical, a 50% chance of sharing one B gene, and only a 25% chance of having completely different B genes. Therefore, when tissue or organ transplants are performed, a deliberate at-

tempt is made to use tissues from siblings or other genetically related, histocompatible people.

Once a tissue donor has been selected, various *in vitro* tests are performed to find how closely the MHCs match. The greater the degree of similarity, the greater the chances are that the transplant will be successful. One such test is the lymphocytotoxic cross-match test. In this test the recipient's serum is mixed with the donor's lymphocytes to determine whether antibodies are present that will cause tissue rejection. The cell-mediated lymphocytosis test, in which recipient leukocytes are mixed with donor serum, is used to test for acute rejection. After a prolonged incubation period, the presence of activated killer cells is an indication that rejection will occur.

Table 32.4 Classes of T Cells

Classes	Function
<i>Class That Kills</i>	
T _C (cytotoxic) cells; also called CD8 ⁺ cells	Cause cytolysis and cell death of target cells—virus-infected cells and tumor cells; important in host defense against cytosolic pathogens
<i>Classes That Regulate Other Cells</i>	
T _H (helper) cells; also called CD4 ⁺ cells	Help B cells make antibody in response to antigenic challenge; stimulate cell-mediated immunity
T _S (suppressor) cells	When mixed with naive or effector T cells, suppress their activity

and acute glomerulonephritis. The effect of MHC molecule expression on the frequency of these diseases is just beginning to be understood.

Types of T Cells

There are several classes or subpopulations of T cells (**table 32.4**). One class (effector) causes cytolysis and cell death of target cells. The other classes regulate specific cells. These effector and regulatory T cells are now discussed in more detail.

Effector Cells

Cytotoxic T (T_C) cells (CD8⁺ cells) are activated in a complex process that involves at least three signals. They attach by T-cell receptor proteins to viral antigen-class I MHC complexes on the surface of virus-infected cells (**figure 32.24**). A CD28-B7 costimulatory signal also is involved. Finally the T_C cells require ex-

posure to interleukin-2 from T_H1 cells. The T_C cells then proliferate and differentiate to form active **cytotoxic T lymphocytes (CTLs)**, which can attack virus-infected cells. CTLs and T_H1 cells release cytokines such as gamma-interferon (IFN-γ), and both macrophages and T cells release tumor necrosis factor-alpha (TNF-α). These cytokines limit virus reproduction, while also activating macrophages and other phagocytes that can destroy the infected cell.

T_H2 cells secrete cytokines such as IL-4, IL-5, and IL-6, which stimulate B-cell proliferation and differentiation into antibody-forming plasma cells (**figure 32.24**). Thus activation of T-helper cells by macrophages is required for the production of both CTLs and antibodies.

Cytotoxic T lymphocytes can destroy target cells in at least two ways. The CD95 pathway triggers apoptosis or programmed cell death. The perforin pathway is probably more often employed by CTLs and uses perforin and granzymes to induce both osmotic cytolysis and apoptosis.

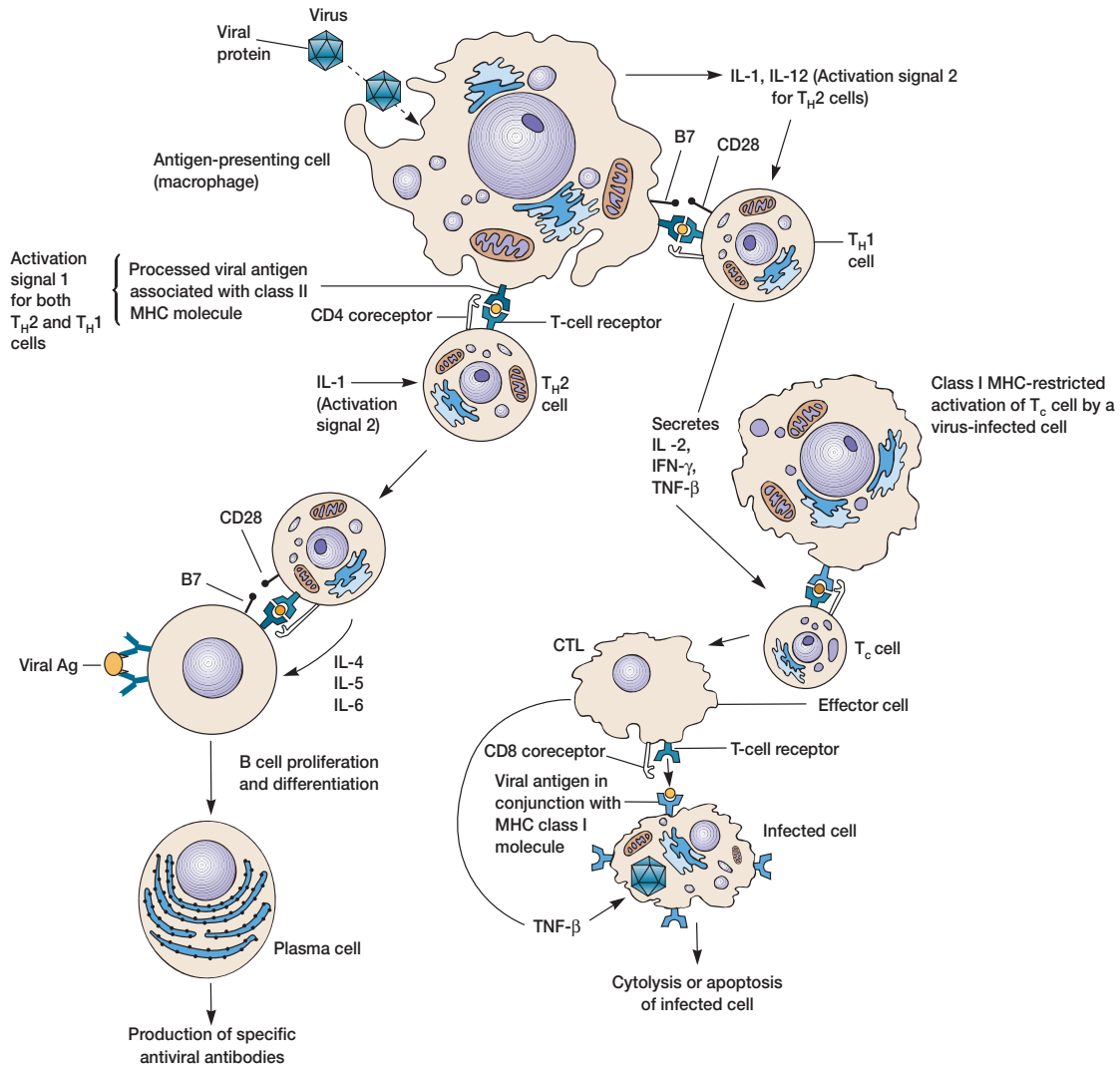
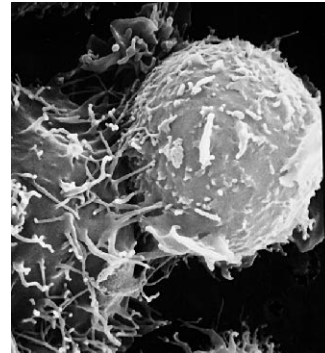
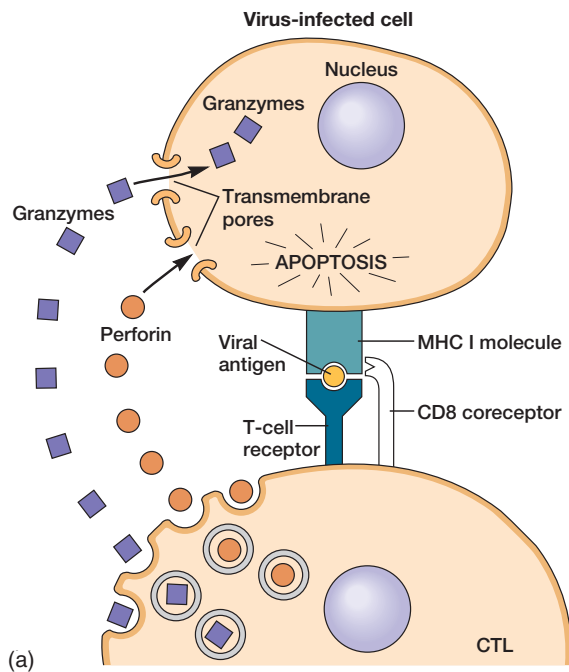
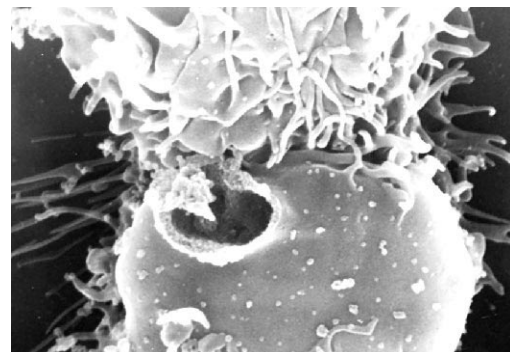


Figure 32.24 Helper T-Cell Responses In Cell-Mediated Immunity. A virus is phagocytosed by a macrophage and a small antigen fragment (peptide) presented to T-helper cells (T_H1 and T_H2) in association with class II MHC molecules. Once activated (activation signals 1 and 2) the T_H2 cell secretes the cytokines IL-4, IL-5, IL-10, and IL-13 that cause B-cell proliferation and secretion of specific antiviral antibodies. Activation signals 1 and 2 also cause the T_H1 cell to secrete IL-2, IFN- γ , and TNF- β . IL-2 regulates the proliferation of cytotoxic T cells. Once the effector T_C cell proliferates, it attacks and causes lysis or programmed cell death (apoptosis) of the virus-infected cell by either the Fas or perforin pathways.



(b)



(c)

Figure 32.25 How an Effector Cytotoxic T Cell Destroys a Virus-Infected Target Cell. (a) T-cell cytotoxicity often involves the perforin pathway and leads to apoptosis and cytolysis. (b) A cytotoxic T cell (left) contacts a target cell (right) ($\times 5,700$). (c) The T cell secretes perforin that forms transmembrane pores in the target cell's plasma membrane. These pores allow the contents of the target cell to leak out (cytolysis).

The **CD95 pathway** depends on the CD95 transmembrane Fas protein receptor, which is found on many eucaryotic cells. CD95 or Fas is coded for by the *fas* gene, a member of the tumor necrosis factor (TNF) family of genes. When the Fas ligand (CD95L) binds to the target cell Fas protein, the CD95-CD95L complex activates several cytoplasmic proteins that initiate a cellular suicide cascade, leading to apoptosis. CTLs have high concentrations of the Fas ligand or CD95L on their surface. The target cell undergoes apoptosis upon binding of the ligand to Fas.

The **perforin pathway** begins when the CTL binds by its T-cell receptor to a viral peptide-class I MHC complex on the target cell surface (figure 32.25a). The following complex series of events now occur:

1. A Ca^{2+} -dependent sequence first takes place: (a) microtubule assembly; (b) movement of cytoplasmic granules toward the

part of the plasma membrane that is in contact with the target cell; (c) reorientation of the Golgi apparatus to the target cell; and (d) movement of the microtubule-organizing center and other cytoskeletal components into the region of the cytoplasm adjacent to the bound target cell.

2. Next the CTL secretes a pore-forming protein, perforin, into the target cell's plasma membrane. Perforin polymerizes in the target cell's membrane to form circular transmembrane pores (figure 32.25c). This can lead to osmotic lysis.
3. Finally the CTL secretes granzymes, proteolytic enzymes that enter the target cell and induce apoptosis.

This sequence of events will cause either cytolysis or apoptosis of the target cell. The CTL then disengages from the dying target cell and recycles its cytoplasmic components in preparation for another attack.

Regulator T Cells

Regulator T cells control the development of effector T cells. Two subpopulations exist: T-helper cells ($CD4^+$ cells) and T-suppressor cells. There are three subsets of **T-helper (T_H) cells**: T_H1 , T_H2 , and T_H0 . The T_H1 and T_H2 subsets produce and secrete a specific mixture of cytokines. **T_H1 cells** produce IL-2, IFN- γ , and TNF- β that are involved in cellular immunity. These cytokines are responsible for delayed-type hypersensitivity reactions and macrophage activation. **T_H2 cells** also produce various cytokines (e.g., IL-4, IL-5, IL-10, IL-13) and are involved in humoral immunity. T_H2 cells are helpers for B-cell antibody responses and defense against helminths. **T_H0 cells** are simply undifferentiated precursors of T_H1 and T_H2 cells.

T-helper cells (T_H1) need two signals (costimulation) to be stimulated by an antigen. The first signal is when antigen fragments are effectively presented to a T cell by a macrophage, dendritic cell, or an activated B cell. For example, when a virus infects a host, some particles may be phagocytosed by a macrophage and partially digested (figure 32.24). Some of the antigenic viral protein fragments are then moved to the surface of the macrophage and presented in close association with class II MHC molecules. This combination of viral antigen fragment and class II MHC molecule is needed for recognition by the T-cell receptor protein on the surface of the T_H1 cell and represents activation signal 1. (The abbreviation CD is from cell-associated differentiation antigen; CDs are functional molecules or receptors that serve as surface markers; section 32.2.)

The second signal for activation involves the CD28 protein receptor on T_H1 cells (**figure 32.26**). When CD28 binds to a protein called B7 (CD80) on the surface of a macrophage (an APC), it delivers a second signal to the T_H1 cell. Both of these signals go into the cytoplasm of the T_H1 cell.

Signal 1 activates a tyrosine kinase located in the cytoplasm. Tyrosine kinase adds phosphate groups to the amino acid tyrosine in proteins. When this phosphorylation occurs, the enzyme phospholipase $C\gamma1$ cleaves the phosphatidylinositol bisphosphate located in the T-helper cell plasma membrane. Two cleavage products are formed and two specific pathways now operate.

One of the cleavage products, diacylglycerol, activates protein kinase C. Protein kinase C moves into the nucleus where it catalyzes the formation of a protein complex called AP-1 from separate components. The other cleavage product, inositol trisphosphate, causes a calcium channel to open; calcium ions to rush into the cytoplasm, leading to further enzymatic activity and the activation of calmodulin, calcineurin, and nuclear factor of activated T_H1 cells (NF-AT). NF-AT then migrates into the nucleoplasm where it binds to the newly formed AP-1, forming a NF-AT/AP-1 complex (a transcription factor). This transcription factor binds tightly to a specific gene sequence in the DNA molecule, causing interleukin-2 mRNA to be transcribed. Interleukin-2 mRNA moves out of the nucleus to the ribosomes where the protein interleukin-2 is produced.

Signal 2 is mediated by the CD28 receptor and B7 molecule. This signal activates another tyrosine kinase, causes the forma-

tion of the transcription factor CD28RC, and also stabilizes the mRNA transcribed from the interleukin-2 gene. This increases the concentration of interleukin-2.

T_H1 cells that are activated by these two signals secrete large amounts of interleukin-2, which activates cytotoxic T cells (figure 32.24). T_H1 cells also secrete gamma-interferon (IFN- γ), which is capable of activating macrophages and increasing microbicidal activity.

The activity of T_H2 cells is depicted in figure 32.24. They are costimulated by antigen presentation and interleukin-1. The T_H2 cells then release several cytokines that stimulate B-cell proliferation and differentiation into antibody-forming plasma cells.

Although the evidence is not conclusive, many immunologists believe in the existence of **T-suppressor (T_S) cells** that can suppress B-cell and T-cell responses. Subpopulations of T_S cells specific to a given antigen can be stimulated to proliferate by the interleukin-2 from activated T-helper cells. This proliferation occurs at a slow rate to provide negative feedback control for that part of the immune response known as acquired immune tolerance.

1. What is the function of an antigen-presenting cell? What is a T-cell receptor and how is it involved in T-cell activation?
2. What is a histocompatibility molecule? What are MHCs and HLAs? Describe the roles of the three MHC classes.
3. Describe antigen processing. How does this process differ for endogenous and exogenous antigens?
4. Briefly describe the cytotoxic T cell, its general role, how it is activated, and the two ways in which it destroys target cells.
5. Outline the functions of a T-helper cell. How do T_H1 and T_H2 cells differ in function? Briefly describe how T_H cells are activated by costimulation.

32.5 B-Cell Biology

Stem cells in the bone marrow produce B cell precursors (*see figure 31.3*). Mature B cells contain cell surface, transmembrane immunoglobulin molecules associated with Ig- α /Ig- β heterodimer proteins; these complexes are called **B-cell antigen receptors (BCRs)**. BCRs can contain IgM and IgD. It is the IgM, however, that is the major receptor for its appropriate antigen. B cells also have receptors for the Fc part of some immunoglobulin classes and may have receptors for complement (C3b). Each B cell may have as many as 50,000 of these membrane-bound immunoglobulins. The total mature B-cell population of each individual thus carries BCRs specific for many antigens; however, each mature B cell possesses BCRs specific only for a particular antigenic determinant.

Antibodies are secreted by plasma cells and bind antigens in solution. The various classes of antibodies (IgG, IgM, IgA, IgD,

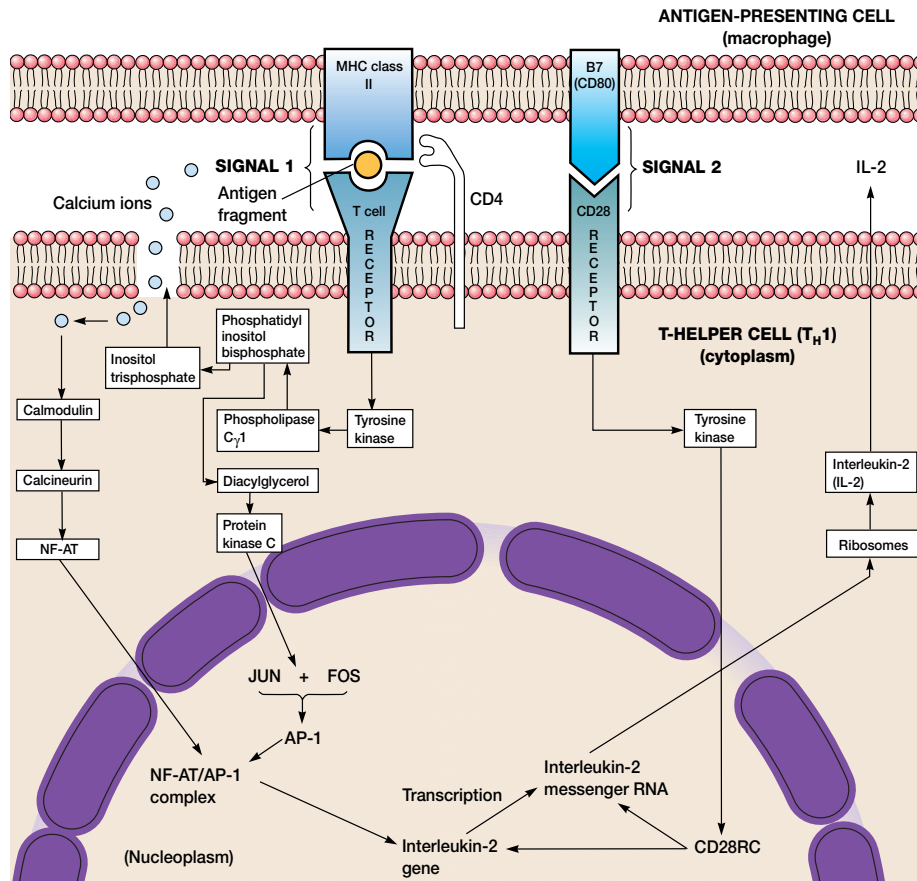


Figure 32.26 Two Signals (Costimulation) Are Essential for T-helper (T_H1) Cell Activation. The first signal is the presentation of the antigen fragment by a macrophage or other antigen-presenting cell along with the MHC class II molecule to the T-helper cell receptor and CD4 protein. The second signal occurs when the macrophage presents the B7 (CD80) protein to the T-helper cell with its CD28 protein receptor. Both signals send information into the cytoplasm of the T-helper cell. The first signal causes interleukin-2 mRNA to be produced. The second signal boosts the production of interleukin-2 to effective concentrations. It now is known that the gene for interleukin-2 is under very tight regulation. It cannot be transcribed unless the NF-AT/AP-1 complex, AP-1, and other transcription factors (e.g., CD28RC) are present. All these factors must be produced anew or activated when the T-helper cell is stimulated through its antigen-specific receptor.

IgE; figures 32.10–32.14) are related to the membrane-bound BCRs. In addition to antigen binding, BCRs also trigger the endocytosis of an antigen leading to it being processed inside the B cell. A small antigen fragment is presented on the surface of the cell in association with major histocompatibility class II molecules. Thus B cells have two immunological roles: (1) they proliferate and differentiate into plasma cells and respond to antigens by making antibodies, and at the same time (2) can act as antigen-presenting cells.

Antigen-Antibody Binding

An antigen binds to an antibody at the antigen-binding site within the Fab region of the antibody. More specifically, a pocket is formed by the folding of the V_H and V_L regions (figure 32.7). At this site, specific amino acids contact the antigen's epitope or haptenic groups and form multiple noncovalent bonds between the antigen and amino acids of the binding site (figure 32.27).

Because binding is due to weak, noncovalent bonds such as hydrogen bonds and electrostatic attractions, the antigen's shape

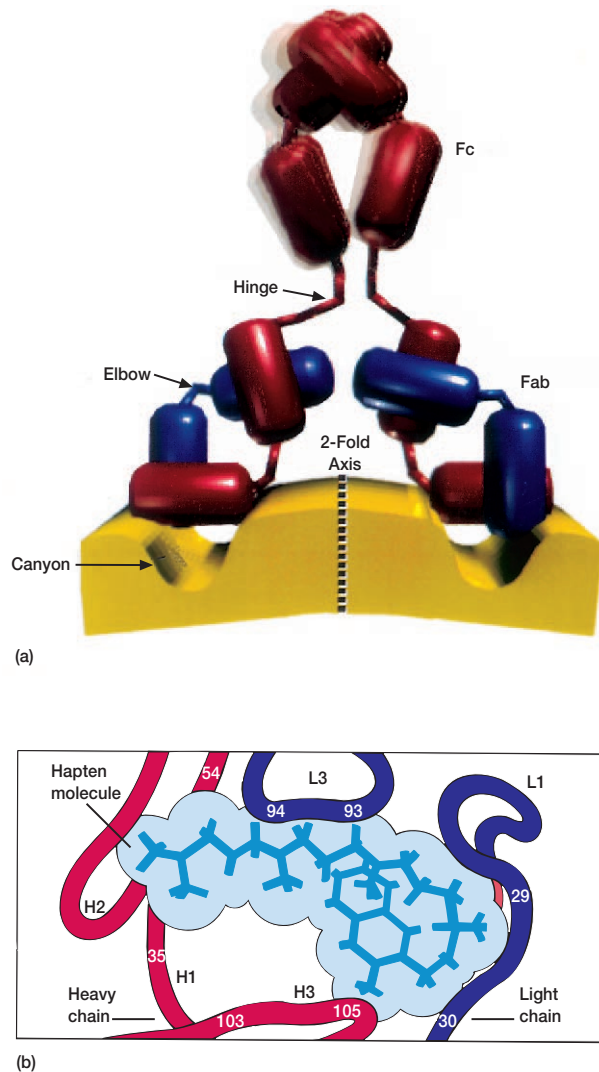


Figure 32.27 Antigen-Antibody Binding. (a) An example of antigen-antibody binding is represented in this model of the monoclonal antibody MAb17-1A bound to the surface of the human rhinovirus. The heavy chains are red, light chains are blue, and the antigen capsid protein is yellow. The RNA interior of the virus would be toward the bottom of the diagram. The antibody is bound bivalently across icosahedral twofold axes of the virus. (b) Based on X-ray crystallography, the hapten molecule nestles in a pocket formed by the antibody combining site. In the illustration the hapten makes contact with only 10–12 amino acids in the hypervariable regions of the light and heavy chains. The numbers represent contact amino acids.

must exactly match that of the antigen-binding site. If the shape of the epitope (figure 32.3) and binding site are not truly complementary, the antibody will not effectively bind the antigen. Although a lock-and-key mechanism normally may operate, in at least one case the antigen-binding site does change shape when it complexes with the antigen (an induced fit mechanism). Regardless of the precise mechanism, antibody specificity results from the nature of antibody-antigen binding.

B-Cell Activation

Upon activation by appropriate signals, the mature B cell proliferates and differentiates into a plasma cell that secretes antibody. Activation is typically antigen specific, activating a particular clone of cells, but can be nonspecific and polyclonal through treatment with B-cell mitogens.

T-Dependent Antigen Triggering

Most antigens have more than one type of antigenic determinant site (epitope) on each molecule (figure 32.3). B cells specific for a given epitope on the antigen (e.g., epitope X) often cannot develop into plasma cells secreting antibody (anti-X) without the collaboration of helper T cells. In other words, binding of the epitope X to the B cell may be necessary, but it is not sufficient for B-cell activation. Antigens that elicit a response with the aid of T-helper cells are called **T-dependent antigens**. Examples include bacteria, foreign red blood cells, certain proteins, and hapten-carrier combinations (figure 32.4).

The basic mechanism for T-dependent antigen triggering of a B cell is illustrated in **figure 32.28** and involves three cells: (1) a macrophage or other antigen-presenting cell to process and present the antigen; (2) a T-helper cell able to recognize the antigen and respond to it; and (3) a B cell specific for the antigen. When all cells and the antigen are present, the following sequence takes place. The macrophage presents part of the antigen and its own class II MHC to the T-helper cell (signal #1 to the T cell). Costimulation is provided by the B7-CD28 interaction (signal #2). Interleukins 1 and 6 stimulate T cells to divide and make interleukins 2 and 4–6. Interleukin-1 also stimulates the hypothalamus to raise the body temperature (causing a fever), which enhances the activity of T cells. Interleukins 2 and 4–6 stimulate T-helper cells to multiply. These helper cells directly associate with B cells that display the correct antigen–class II MHC complex and secrete cytokines called B-cell growth factors (BCGFs), which cause B cells to multiply. As the number of B cells increase, T-helper cells produce other cytokines, the B-cell differentiation factors (BCDFs). These instruct some B cells to stop replicating, to differentiate into plasma cells, and to start producing antibodies. B-cell activation involves more than T-helper cell activity. The B cell also recognizes the antigen through its surface IgM receptors for the antigen (signal #1 for the B cell) and is subsequently triggered to proliferate and differentiate into plasma cells secreting antibody. (One plasma cell can synthesize more than 10 million antibody molecules per hour!) It should be noted here

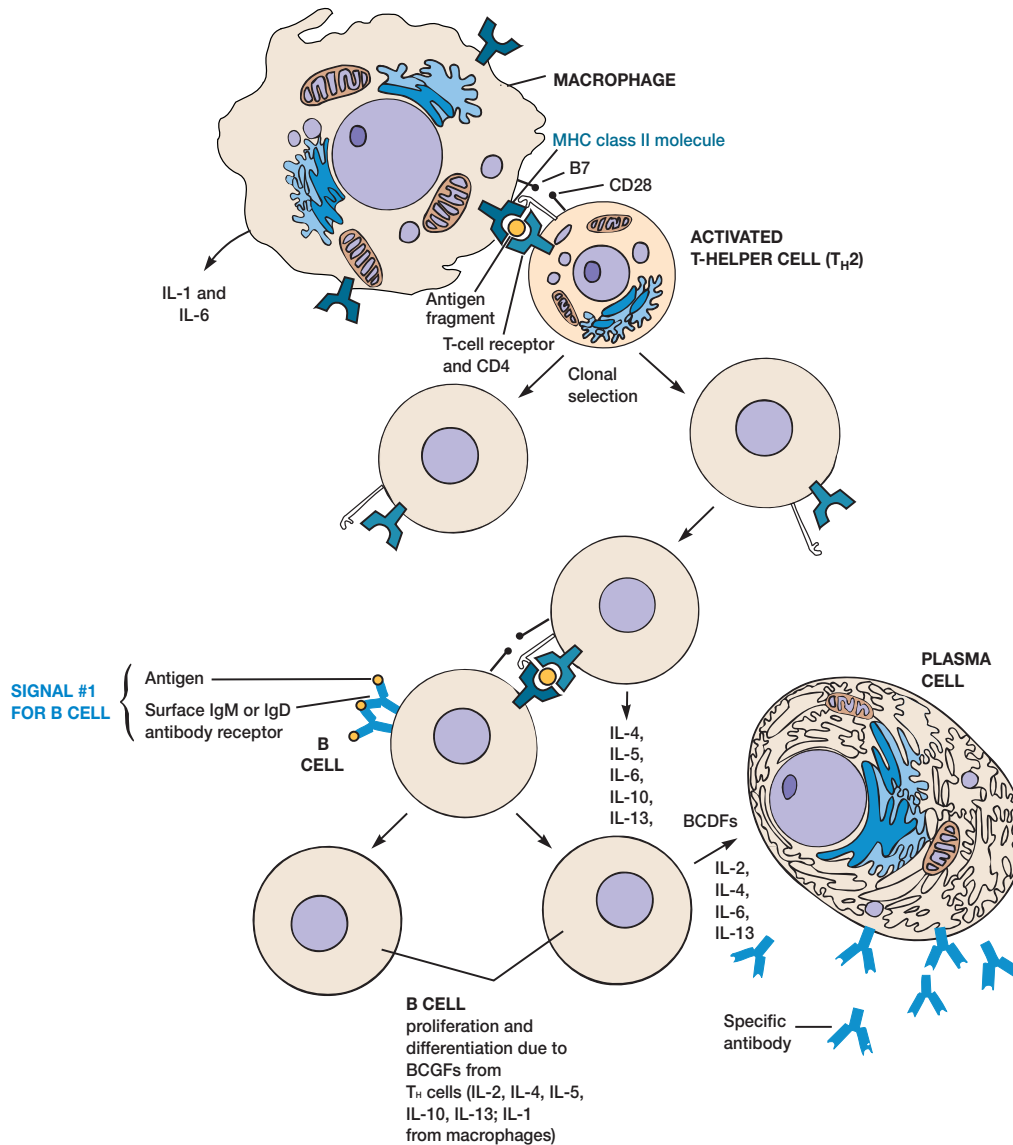


Figure 32.28 T-Dependent Antigen Triggering of a B Cell. Schematic diagram of the events occurring in the interactions of macrophages, T-helper cells, and B cells that produce humoral immunity. Many cytokines (e.g., IL-1, IL-4, IL-5, IL-10, IL-13) stimulate B-cell proliferation and may be called B-cell growth factors (BCGFs). Cytokines such as IL-2, IL-4, IL-6, and IL-13 stimulate B-cell differentiation and are called B-cell differentiation factors (BCDFs).

that B cells are more effective antigen presenters than macrophages at low antigen concentrations if the antigen can bind to surface immunoglobulins. The B cells bind antigen, take it up by receptor-mediated endocytosis, and present it to T-helper cells to activate them. In this situation B cells and T_H cells activate each other. All the immune responses that produce IgG, IgA, and IgE involve this T-dependent antigen triggering. [Fever \(pp. 722–23\)](#)

T-Independent Antigen Triggering

Not all antibody responses require T-cell help. There are antigens that trigger B cells into immunoglobulin production without T-cell cooperation. These are called **T-independent antigens**. Examples include certain tumor-promoting agents, anti-immunoglobulin (anti-Ig), antibodies to certain B-cell differentiation antigens, and

Table 32.5 Comparison of Lymphocytes Involved in the Immune Response

Property	T Cells	B Cells
Origin	Bone marrow in adults	Bone marrow in adults
Maturation and differentiation	Thymus	Lymphoid tissue; bursa of Fabricius in birds
Mobility	Great	Very little (some stages circulate)
Complement receptors	Absent	Present
Surface immunoglobulins	Absent	Present
Proliferation	Upon antigenic stimulation, differentiate into effector and memory cells	Upon antigenic stimulation, proliferate and differentiate into plasma and memory cells
Immunity type	Cell mediated and humoral (B cell activation by T _H cells)	Humoral
Distribution	High in blood, lymph, and lymphoid tissue	High in spleen, lymph nodes, bone marrow, and other lymphoid tissue; low in blood
Secretory product	Cytokines	Antibodies
Subsets and functions	T-helper (T _H) cell: necessary for B-cell activation by T-dependent antigens and T-effector cells. There are three types of T-helper cells: T _H 1, T _H 2, and T _H 0. T-suppressor (T _S) cell: blocks induction and/or activation of T _H cells and B cells; helps maintain tolerance Cytotoxic T (T _C) cell: differentiates into a CTL that lyses cells recognized as nonself and virus or parasite-infected cells Memory cells	Plasma cell: a cell arising from a B cell that manufactures specific antibodies Memory cell: a long-lived cell responsible for the anamnestic response

Table 32.6 Antigen Recognition by T and B Cells

Characteristic	T Cells	B Cells
Binds soluble antigen	No	Yes
Biochemistry of the antigens	Mostly proteins, but some glycolipids presented on MHC molecules	Proteins, glycolipids, polysaccharides
Antigen recognition	Antigens processed internally and presented as linear peptides bound to MHC molecules Involves three partners: T-cell receptor, antigen, and MHC molecule	Accessible areas of protein structure containing sequential amino acids and nonsequential amino acids Immunoglobulin receptor binds antigen in native conformation Involves two partners: antigen and membrane immunoglobulin

bacterial lipopolysaccharides. The T-independent antigens are polymeric—that is, composed of repeating polysaccharide or protein subunits. They elicit almost exclusively IgM antibody formation, and there is little switching to IgG; the resulting antibody generally has a low affinity for antigen.

The mechanism for activation by T-independent antigens probably depends on their polymeric structure. These large molecules present a large array of identical epitopes to a B cell specific for that determinant. The multivalent nature of the cell receptor (surface IgM) is such that cell activation occurs and IgM is secreted. Because there is no T-cell help, the B cell cannot switch to high-affinity IgG production, and no memory B cells are formed.

Section 32.4 presented the biology of T-cells and this current section describes B-cell biology. **Tables 32.5** and **32.6** summarize and compare many of the important properties of lymphocytes that were discussed in these two sections.

1. What are B-cell antigen receptors and how are they involved in B-cell activation?
2. Briefly compare and contrast B cells and T cells with respect to their formation, structure, and roles in the immune response.
3. How does antigen-antibody binding occur? What is the basis for antibody specificity?
4. How does T-independent antigen triggering of B cells differ from T-dependent triggering?

32.6 Action of Antibodies

The antigen-antibody interaction is a bimolecular association that exhibits exquisite specificity. The *in vivo* interactions that occur in vertebrate animals are absolutely essential in protecting the animal against the continuous onslaught of viruses, microorganisms and their products, certain macromolecules, and cancer cells. Outside the animal body (*in vitro*), this same specificity has led to the development of a variety of immunological assays that can detect the presence of either antibody or antigen. They are important in the diagnosis of diseases; in the identification of specific viruses, bacteria, and parasites; in monitoring the level of the humoral response and immunologic problems; and in identifying molecules of medical and biological interest. Immunological assays differ in their speed and sensitivity; some are qualitative whereas others are quantitative. The rest of this section examines those antigen-antibody interactions that occur in the animal body (*in vivo*). Antigen-antibody reactions that are routinely dealt with in the laboratory (*in vitro*) are covered in chapter 33, which is concerned with medical immunology.

Toxin Neutralization

Immunity to a disease like diphtheria depends on the production of specific antibodies that inactivate the toxins produced by the bacteria. This process is termed **toxin neutralization** (figure 32.29). Once neutralized, the toxin-antibody complex is either unable to attach to receptor sites on host target cells, unable to enter the cell, or it is ingested by macrophages. For example, diphtheria toxin inhibits protein synthesis after binding to the cell surface by the B fragment and subsequent passage of the active A fragment into the cytoplasm of the target cell (see figure 34.5). Thus the antibody blocks the toxic effect by inhibiting the entry of the A fragment or the binding of the B fragment. An antibody capable of neutralizing a toxin or antiserum containing neutralizing antibody against a toxin is called **antitoxin** (figure 32.29b). [Exotoxins \(pp. 794–99\)](#)

Viral Neutralization

IgG, IgM, and IgA antibodies can bind to some viruses during their extracellular phase and inactivate them. This antibody-mediated viral inactivation is called **viral neutralization**. Fixation of the classical pathway (section 32.7) complement component C3b to the virus aids the neutralization process. Viral neutralization prevents a viral infection due to the inability of the virus to bind to its target cell (figure 32.29c).

Adherence Inhibition

The capacity of bacteria to colonize the mucosal surfaces of mammalian hosts is dependent in part on their ability to adhere to mucosal epithelial cells. Secretory IgA (sIgA; pp. 736–38) antibodies inhibit certain bacterial adherence promoting factors. Thus sIgA has a unique role in protecting the host against infection by some pathogenic bacteria, and perhaps by other microorganisms on mucosal surfaces.

IgE and Parasitic Infections

Immune reactions against protozoan and helminthic parasites are only partially understood. Parasites that have a tissue invasive phase in their life cycle often are associated with both eosinophilia (an excessive number of eosinophils in the blood) and elevated IgE levels. Recent evidence shows that, in the presence of elevated IgE, eosinophils can bind to the parasites and discharge their granules. Degranulation releases lytic and inflammatory mediators that destroy the parasites (see figure 33.2).

Opsonization

Phagocytes have an intrinsic ability to bind directly to microorganisms by nonspecific cell surface receptors, engulf the microorganisms, form phagosomes, and digest the microorganisms (see figure 31.16). This phagocytic process can be greatly enhanced by opsonization. As noted in section 31.8, opsonization is the process in which microorganisms or other particles are coated by antibody and/or complement, and thus prepared for “recognition” and ingestion by phagocytic cells. Opsonizing antibodies, especially IgG1 and IgG3, bind to Fc receptors on the surface of macrophages and neutrophils. This binding forms a bridge between the phagocyte and the antigen. [Phagocytosis \(pp. 718–20\)](#)

Immune Complex Formation

Because antibodies have at least two antigen-binding sites and most antigens have at least two antigenic determinants, cross-linking can occur, producing large aggregates termed **immune complexes** (figure 32.30). If the antigens are soluble molecules and the complex becomes large enough to settle out of solution, a **precipitation** [Latin *praecipitare*, to cast down] or **precipitin reaction** has occurred, and is caused by a **precipitin** antibody. When the immune complex involves the cross-linking of cells or particles, an **agglutination reaction** occurs and the responsible antibody is an **agglutinin**. Agglutination specifically involving red blood cells is a **hemagglutination** reaction and is caused by a **hemagglutinin**. These immune complexes are more rapidly phagocytosed *in vivo* than are free antigens.

The extent of immune complex formation, whether within an animal or *in vitro*, depends on the relative concentrations of the precipitin antibody and antigen. If there is a large excess of antibody, separate antibody molecules usually bind to each antigenic determinant and a less insoluble network or lattice forms (see figure 33.18a). When antigen is present in excess, two different antigen molecules tend to bind to each antibody and network development or crosslinking also is inhibited. In the equivalence zone the ratio of antibody and antigen concentrations is optimal for the formation of a large network of interconnected antibody and antigen molecules. All antibody and antigen molecules precipitate or agglutinate as an insoluble complex. Precipitin reactions can occur in both solutions and agar gel media. In either case, antibody-antigen equivalence is required for optimal results.